

Review

Plant diseases recognition on images using convolutional neural networks: A systematic review

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ABSTRACT

Plant diseases are considered one of the main factors influencing food production and minimize losses in production, and it is essential that crop diseases have fast detection and recognition. The recent expansion of deep learning methods has found its application in plant disease detection, offering a robust tool with highly accurate results. In this context, this work presents a systematic review of the literature that aims to identify the state of the art of the use of convolutional neural networks (CNN) in the process of identification and classification of plant diseases, delimiting trends, and indicating gaps. In this sense, we review 121 papers selected in the last ten years with different approaches to treat aspects related to disease detection, characteristics of the dataset, the crops and pathogens investigated. From the results of the systematic review, it is possible to understand the innovative trends regarding the use of CNN's in the identification of plant diseases and to identify the gaps that need the attention of the research community.

1. Introduction

Plant diseases are considered one of the main factors influencing food production, being responsible for the significant reduction of the physical or economic productivity of the crops and, in some instances, maybe an impediment to this activity. According to Altieri (2018), to minimize production losses and maintain crop sustainability, it is essential that disease management and control measures be carried out appropriately, highlighting the constant monitoring of the crop, combined with the rapid and accurate diagnosis of the diseases. These practices are the most recommended by phytopathologists (Altieri, 2018).

The major challenge for agriculture is the correct identification of the symptoms of major diseases that affect crops (Mohanty et al., 2016). Manual and mechanized practices in traditional planting processes cannot cover large areas of plantations and provide essential early information to decision-making processes, according to Miller et al. (2009). Thus, it is necessary to develop automated solutions, practical, reliable, and economically able to monitor the health of plants providing meaningful information to the decision-making process, for example, the application and correct dosage of pesticides in specific treatment certain diseases (Mahlein, 2016).

Computer Vision, along with Artificial Intelligence (AI), has been developing techniques and methods for recognizing and classifying objects with significant advances (Arnal Barbedo, 2013). According to LeCun et al. (2015) and, deep learning allows computational models to learn representations of data with multiple levels of abstraction, improving the state-of-the-art in many domains, such as speech recognition, object recognition, object detection.

The simplest Deep Learning models are called Deep Feedforward, in which information is only propagated in one direction through neurons. Other examples of algorithms are: Back-Propagation, Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), including Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), Auto-Encoder (AE), Deep Belief Network (DBN), Generative Adversarial Network (GAN), and Deep Reinforcement Learning (DRL) (Goodfellow et al., 2016). One particular type of deep, feedforward network that was much easier to train and generalized much better than networks with full connectivity was the convolutional neural networks (CNN's) (Salas et al., 2019).

CNN's constitute one of the most powerful techniques for modeling complex processes and performing pattern recognition in applications with a large amount of data, as the pattern recognition in images (LeCun et al., 2015). This is a connectionist approach, stands out as one of the

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most prominent because it allows the automatic extraction of features, and their results in some experiments are already superior to humans in large-scale reconnaissance tasks.

Many approaches already make use of popular architectures such as LeNet (Lecun et al., 1998), AlexNet (Krizhevsky et al., 2012), VGGNet (Simonyan and Zisserman, 2014), GoogLeNet (Szegedy et al., 2015a), InceptionV3 (Szegedy et al., 2015b), ResNet (He et al., 2016) and DenseNet (Huang et al., 2017), considerably increasing the accuracy in the recognition and identification of plant diseases. The works of Tian et al. (2019), Liang et al. (2019), Long and Liu (2019) and Abade et al. (2019) highlight the use of images as one of the most advanced methods of detection and recognition of plant diseases.

Despite the significant advances provided by deep learning, numerous challenges still hinder the recognition and identification of plant diseases. Factors such as the genetic and phenotypic diversity of crops and the great diversity of pests and diseases often detract from the results achieved when applied in real cultivation environments. Additionally, it should be noted that the characteristics of the data sets are limited to the identification and recognition of diseases caused by biotic factors. However, there is a similarity in the characteristics caused by abiotic factors, such as nutrient deficiency, soil compaction, water stress, among others that can also detract from the results, leading to flawed diagnoses.

Thus, this systematic literature review (SLR) is intended to characterize the state of the art recognition and identification of plant diseases using convolutional neural networks. The investigation sought significant contributions regarding the challenges listed and the different innovations that aim to improve the performance of CNN's and, consequently the correct identification of diseases. A systematic literature review is a type of scientific research that aims to present an unbiased report about a research topic, following a methodology that is reliable, accurate and that allows auditing (Kitchenham, 2004; Fabbri et al., 2013).

The remainder of the paper is organized as follows: In Section 2 presented the background. Section 3 describes our review protocol in detail. Section 4 presents the results of the SLR. Section 5 presents the discussion and finally Section 6 concludes the paper.

2. Background

The identification of plant diseases is one of the most basic and essential agricultural activities. In most cases, identification is performed manually, visually, by serological and molecular tests or by microscopy. The problem with the visual assessment to identify diseases is that the evaluating subject has assumed a subjective task, prone to psychological and cognitive phenomena that can lead to prejudice, optical illusions, and, finally, error (Barbedo, 2016).

2.1. Types of plant diseases

Plant diseases are classified according to the type of their causative agent. Diseases originating from living organisms are called biotic, unlike diseases that are produced by non-living ecological circumstances, called abiotic (Husin et al., 2012). Fungi, bacteria, viruses, plague, nematodes are the leading causes of different forms of biotic diseases (Agrios, 2005). In this study, the pest or plague, or more specifically, biological pest, is the outbreak of certain species harmful to agricultural development that cause epidemic diseases in plants and their cultivars.

2.2. Plant disease detection system

The automated solutions for the identification of plant diseases using images and machine learning, especially CNN's, have provided significant advances to maximize the accuracy of correct diagnosis. Convolutional Neural Networks is a subset of machine learning approaches that have emerged as a versatile tool for assimilating large amounts of

heterogeneous data and providing reliable predictions of complex and uncertain phenomena (Goodfellow et al., 2016).

2.3. Evolution of CNN architectures

Different innovations in CNN architectures have been proposed since 1998 with the presentation of LeNet-5 (Lecun et al., 1998). It can be categorized as parameter optimization, regularization, structural reformulation, etc. However, it is observed that the main improvement in the performance of CNN's is motivated by the fundamental reformulation and design of new blocks (Khan et al., 2019). Thus, this SLR classifies as new architecture that is by the innovation proposals reported in studies of Khan et al. (2019) and Goodfellow et al. (2016).

The state of the art of convolutional neural network architecture is organized to solve tasks of classification, segmentation, and object detection. In the models for classification problems, the input images are encoded in different steps with convolution and subsampling layers, and finally, the classifier approach that used to calculate class probability. However, these models with the classification layer can be used as feature extraction for segmentation and detection tasks (Alom et al., 2019).

There are several semantic segmentation models that have been proposed in the last few years. The segmentation model consists of two units: Encoding and decoding units (Alom et al., 2019). The very first segmentation model is Fully Convolutional Network (FCN) proposed by Long et al. (2015). There are several new models have recently proposed which includes UNet (Ronneberger et al., 2015), SegNet (Badrinarayanan et al., 2017), DeepLab (Chen et al., 2018), and R2U-Net (Alom et al., 2019).

The detection problem is a bit different compared to classification and segmentation problems. The model's goal is to identify target types with their corresponding position. The model answers two questions: What is the object (classification problem)? And where the object in the image (regression problem)? The most popular models for object detection are Region-based CNN (R-CNN) (Girshick et al., 2014), You only look once (YOLO) (Redmon et al., 2016), SSD: Single Shot MultiBox Detector (Liu et al., 2016), Focal Loss for Dense Object Detection (Lin et al., 2017), Faster R-CNN (Ren et al., 2017), Fast R-CNN (Wang et al., 2017), Mask R-CNN (He et al., 2017), and YOLACT: Real-time Instance Segmentation (Bolya et al., 2019).

It is important to note that although CNN's architectures are powerful in object classification, segmentation, and detection tasks, they require a large amount of data to be trained with, given the number of parameters and deep layers to convey generalization. This need for large amounts of data is often mitigated employing methods such as Transferring Learning (Shao et al., 2015), Data Augmentation (Shorten and Khoshgoftaar, 2019), Customizing Layers (Sledevic, 2019) and Optimizing Hyperparameters (Diaz et al., 2017).

3. Planning review

The planning of this SLR followed the processes defined by Fabbri et al. (2013). To provide transparency and allow the replication and auditing of the entire process (essential requirements in systematic reviews), the StArt Tool (State of the Art Through Systematic Review) was used (Hernandes et al., 2012), which supports all phases of an SLR.

In the planning phase, we need to identify the real need, namely the motivation for implementing the SLR. Thus, defining the objectives and elaborating a protocol for SLR are essential elements for successful execution of the technique. It should be noted that the quality of the protocol directly impacts the SLR results.

3.1. Research questions

It is known that CNN's have been showing better results than traditional Computer Vision techniques and other techniques using

Table 1
Selection criteria for studies defined in the SLR planning protocol.

ID	Criterion Type		Description
	Inclusion	Exclusion	
C1	X	–	Studies that approaches the identification of plant diseases through convolutional neural networks.
C2	X	–	Studies that present new models or architectures of convolutional networks.
C3	X	–	Studies describing techniques or methods for identifying plant diseases using CNN's.
C4	–	X	Studies without full text available
C5	–	X	Studies not written in English
C6	–	X	Duplicate publication from multiple sources
C7	–	X	Studies without qualifying data for extraction
C8	–	X	Studies that do not use CNN's as the main approach

Machine Learning. Thus, this RSL is intended to identify the characteristics that affect the efficiency of plant disease identification approaches using convolutional neural networks. And the main research question of SRL that we intend to answer is:

"How are convolutional neural networks innovating and overcoming the challenges of the plant disease identification process?"

Additionally, secondary questions were formulated to assist in the identification of characteristics that determine the desired response, namely:

- SQ1 – Which approaches make use of new architectures or models?
 SQ2 – What are the characteristics of the datasets predominantly used?
 SQ3 – What types of crops are most investigated with approaches using CNN's?
 SQ4 – What types of approaches and frameworks are commonly used?
 SQ5 – Which CNN algorithm is prevalent in current approaches?
 SQ6 – What types of plant diseases are most investigated with approaches using CNN's?

3.2. Search strategy

This SLR focuses its research on digital scientific databases, so any type of gray literature was not included in the investigation process. The study is based on the assumption that most of the research with relevant results that may appear in the gray literature are typically described or referenced in scientific papers already published.

The search and selection strategy for primary studies focused on 5 indexed electronic databases, namely: ACM Library ¹, IEEE Xplore Digital Library ², Elsevier Scopus ³, Springer Link ⁴ and Google Scholar ⁵. The databases were selected because they offer the largest volume of high-impact scientific conference proceedings and journals, broadly covering the field of investigations on plant diseases and machine learning through CNN's.

According to Petersen et al. (2008) and Fabbri et al. (2013), the process of collecting the study set begins, by defining the *Search String*. Thus, the *Search String* should be formulated based on the experience of the researchers and reviewers involved in the SLR process. Even though different indexed research repositories define their syntax for building search strings, we initially defined the following generic string that was later customized for each search engine:

("plant disease" OR "plant pathology" OR "crop disease")
 AND

Table 2
Criteria for Quality Assessment of studies defined in the SLR planning protocol.

ID	Question	Indicator			Reference
		Y	P	N	
Q1	Are the studies cited by other works and scientific investigations?	1,0	0,5	0,0	Yes ⇒ Studies published studies with a significant number of citations. Partially ⇒ Studies published with a low number of citations Not ⇒ Studies published without citations
Q2	Do the studies describe the approaches in detail and provide subsidies for replication and finding results?	1,0	0,5	0,0	Yes ⇒ Studies explicitly describe the approaches; Partially ⇒ Studies describe the approaches, however, they do not provide subsidies for replication and verification of results Not ⇒ Studies that do not provide sufficient characteristics to describe the approaches, nor do they allow replication and auditing;
Q3	Are the main findings stated clearly? regarding credibility, validity, and reliability?	1,0	0,5	0,0	Yes ⇒ The findings are clearly reported Partially ⇒ The main findings are partially reliable and credible. Not ⇒ There is no credibility in the findings presented.
Q4	Are the limitations of the studies clearly specified and documented?	1,0	0,5	0,0	Yes ⇒ Specifies and documents study limitations Partially ⇒ It mentions the limitations, however it does not specify or document its determinants. Not ⇒ It does not mention the limitations and threats of validity of the studies.
Q5	Are the conclusions credible and consistent with the results presented?	1,0	0,5	0,0	Yes ⇒ Methodologically prepared studies with consistency of facts and confidence in the discoveries presented Partially ⇒ Methodologically elaborated studies, but with insufficient details to certify the results and conclusions. Not ⇒ Studies methodologically poorly elaborated and without credibility.

("machine learning" OR "convolutional neural network" OR "deep learning" OR "DNN" OR "CNN")

3.3. Study selection criteria

The identification and recognition of plant diseases through convolutional neural networks has aroused a high interest in the scientific

¹ <https://dl.acm.org> access:January/2020

² <https://ieeexplore.ieee.org/Xplore> access:January/2020

³ <https://www.scopus.com> access:January/2020

⁴ <https://link.springer.com> access:January/2020

⁵ <https://www.scholar.google.com> access:January/2020

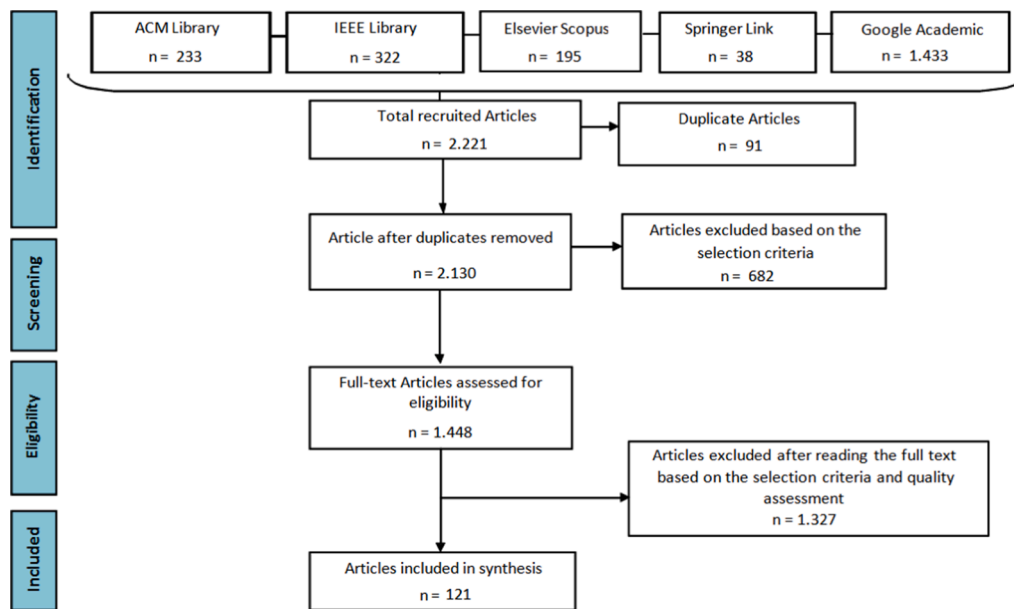


Fig. 1. Flowchart of study phases and selections – adapted from Liberati et al. (2009).

community in the last decade. Therefore, publications from the previous ten years (from 2010 to 2019) were selected.

The search string had a broad scope intentionally because we did not want to miss any potentially exciting research. This full scope has led to a large number of papers, from which we filtered the most relevant ones using the selection criteria presented in Table 1.

Once identified, studies need to be selected through the application of selection criteria (which are the inclusion and exclusion criteria) and can be evaluated by the quality criteria. The selection criteria must specify the main characteristics and/or content that the studies must have to be included or excluded.

3.4. Study quality assessment

The quality criteria aim to evaluate methodological aspects of the studies, that is, they can be considered issues such as the relevance of the research theme and the use of methods that lead to the objectives proposed in the study.

The quality assessment is performed using forms developed by the

researchers. These are composed of quality criteria, that is, questions about methodological aspects of each study (Kitchenham, 2012; Staples and Niazi, 2007). Table 2 presents the items that make up the study quality assessment form.

We chose the criteria based on their influence on the quality of the final product. Points were given to each of the five criteria based on the following scale: yes = 1, partially = 0.5, not = 0.

The quality of each study is directly associated with its methodological quality, which can be measured by aspects such as internal validity (effectively measuring what you want to measure), external validity (ability to generalize the results), the relevance of the research topic (questions research and well-defined objectives and based on the literature), adoption of methods that lead to the proposed goals, among others.

Primary studies that obtained a quality index (IQ) of less than 3 points were excluded from this SLR, to guarantee the high quality of the summarized final set.

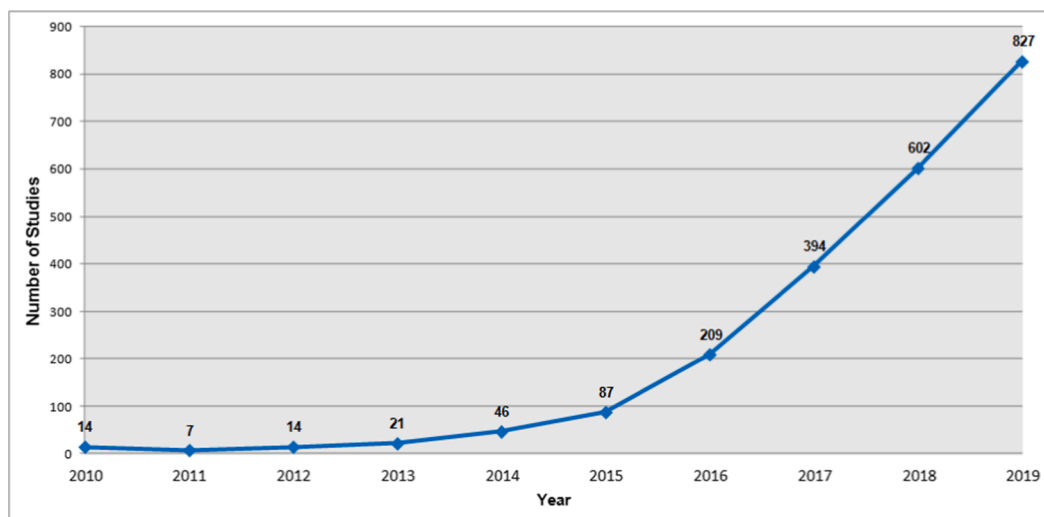


Fig. 2. Temporal evolution of the quantity of publications recruited containing the keywords defined in the Search String in the last ten years (2010 to 2019).

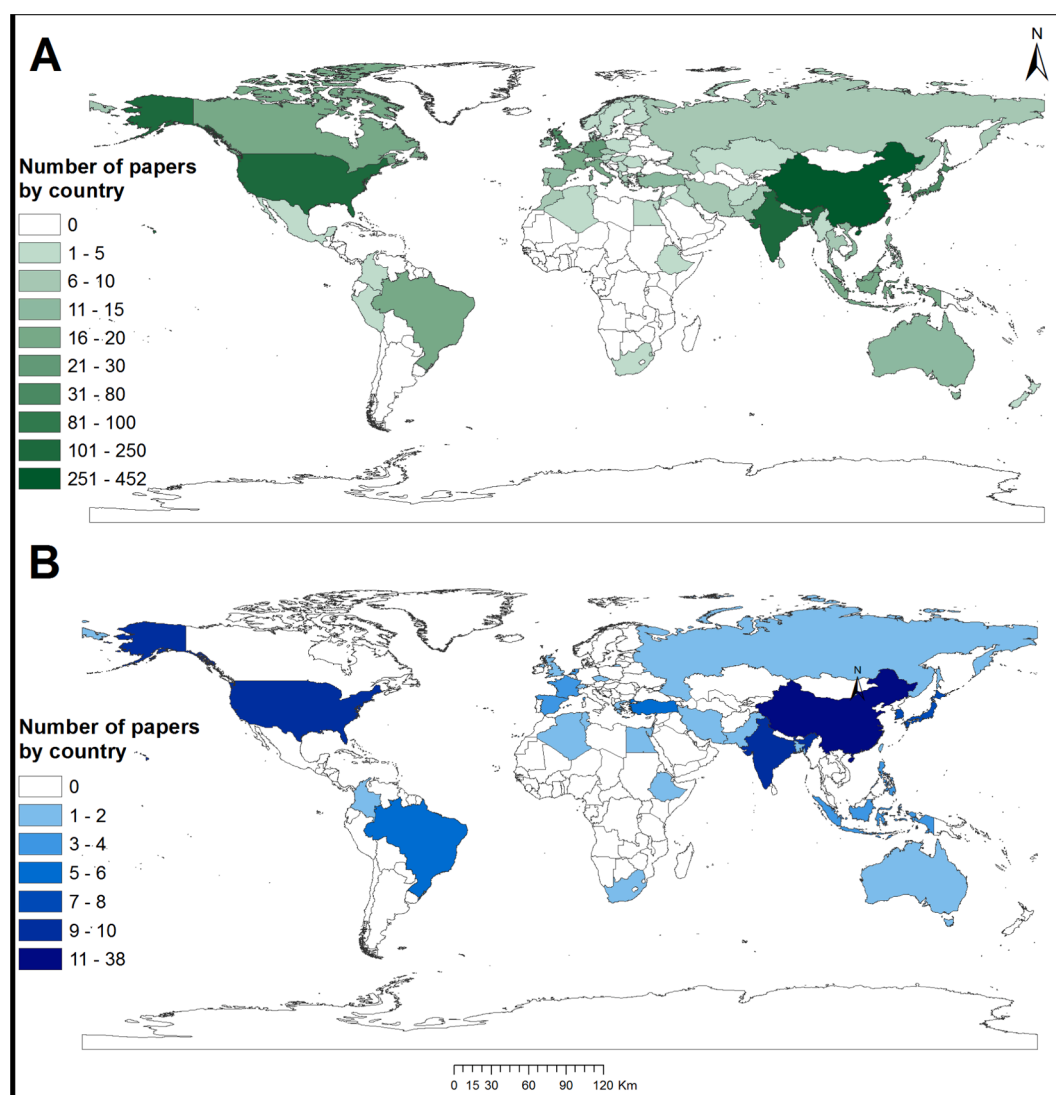


Fig. 3. (A) Geographical distribution of the studies assessed for eligibility and (B) the studies included in the final selection.

3.5. Data extraction and synthesis

To elect the primary studies, we use the PRISMA (*Preferred Reporting Items for Systematic and Meta-Analyses*) recommendation, according to its four stages: identification, screening, extraction, and final selection (Liberati et al., 2009). In the identification, the study sample was extracted, quantified, and stored eliminating duplicate studies. In the screening stage, the reviews were pre-selected from reading the title, abstract, and keywords by two independent reviewers, obtaining a Kappa index $> 80\%$ ($p < 0.05$) of interobserver agreement. Discrepancies were resolved by a third reviewer (Buscemi et al., 2006).

In the extraction stage, the full reading of pre-selected studies was performed, excluding studies that did not meet the inclusion criteria, defining the final sample. The synthesis activity aims to combine the data extracted from each of the selected primary studies, delineating the situation of the investigated research questions.

A total of 2,221 studies were retrieved, where 1,433 were identified in the Google Scholar, 322 in the IEEE Library, 233 in the ACM Library database, 195 in the Elsevier Scopus, and 38 in the Springer Link.

During the identification phase, 91 duplicate studies were excluded, and during the initial selection, 682 reviews were eliminated because

they did not meet the established inclusion criteria. For the extraction phase, 1,448 reviews were selected, and, after reading the entire text, 121 reviews were summarized. Fig. 1 illustrates the steps followed in conducting the SLR with a summary of the results.

The investigation was carried out retrieving studies between the years 2010 to 2019, it is possible to note that the year 2012 is a milestone not only for studies that address convolutional networks, but also the application of this approach to the identification and recognition of plant diseases. Fig. 2 presents the evolution of studies that address this theme.

Additionally, the studies identified were distributed geographically, as illustrated by Fig. 3. Fig. 3 (A) shows the geographical distribution of 1,448 studies eligible for data extraction and synthesis. In Fig. 3 (B), we present the distribution of 121 studies included in the synthesis of the results of this SLR.

It was noted that the investigations and consequently the scientific interest on the identification and recognition of plant diseases using machine learning techniques are coincident with the most populous countries in the world, among them: China, India, the United States of America, Indonesia, Pakistan, and Brazil.

At the end of the synthesis activities, Table A.3 was prepared,

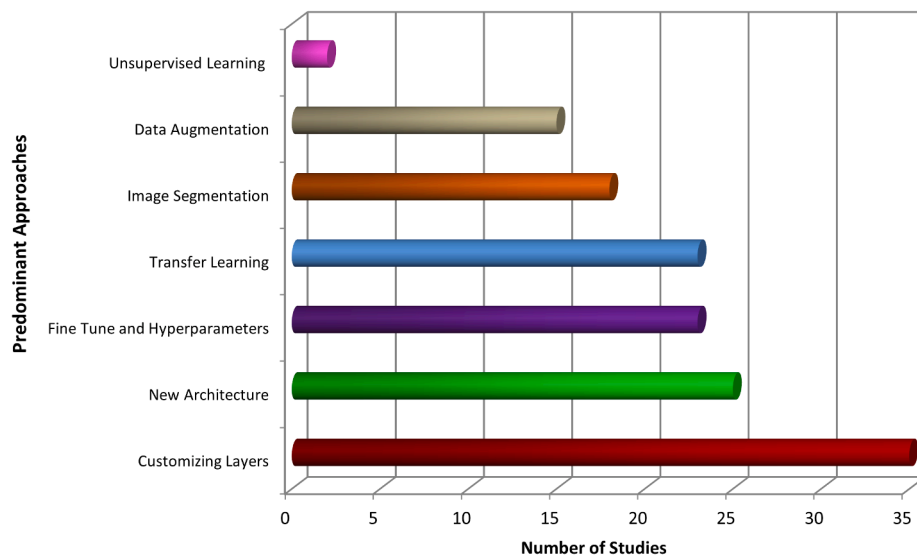


Fig. 4. Predominant Approaches in 121 Summary Studies.

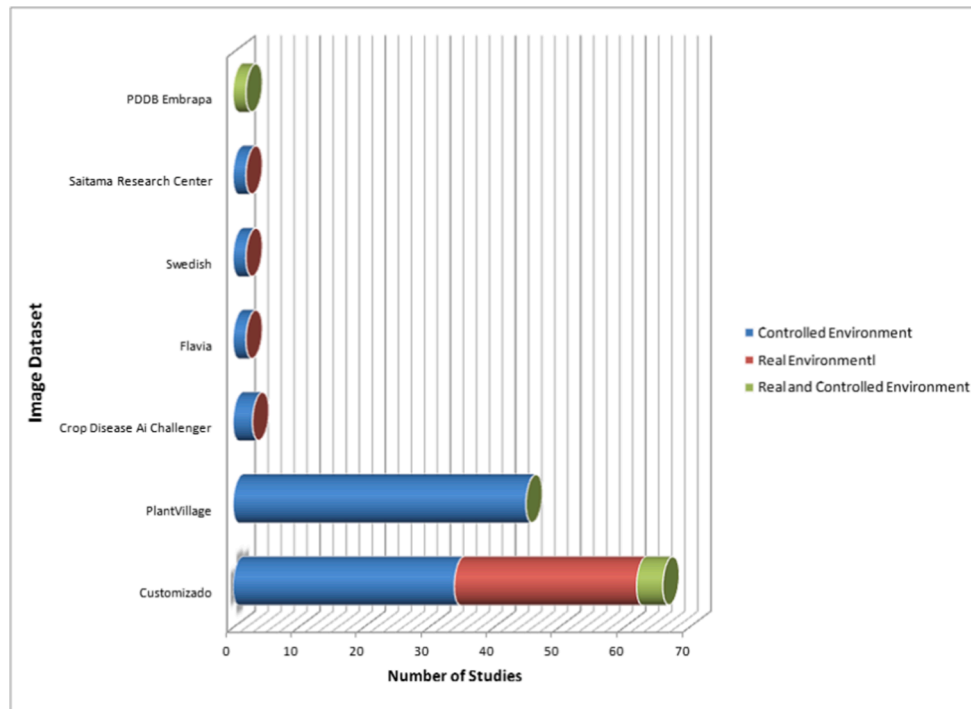


Fig. 5. Characteristics of the datasets present in the 121 Summary Studies.

available in Appendix A with 121 primary studies chosen for extraction. Basically, the table is composed by the following attributes: ID (Study identification), Refs (Author of the approach), IQ (Quality Index), Algorithm, Best accuracy, Dataset characteristics, Crop type, Image type, Language and Framework, Predominant approach. In addition, in Table B.4, information about each investigated disease was detailed according to the type of crops and causative pathogen. Information from Table B.4 is available in Appendix B.

Each column of the Table summarizes a relevant feature of the investigation process that is directly related to the questions formulated for the SLR.

4. Results

In this section, we present the discrete and continuous quantitative data extracted from the 121 primary studies that support formulating the answers corresponding to the six secondary questions of the research.

4.1. SQ1: Which approaches make use of new architectures or models?

To answer this question, the data extraction from the 121 summarized studies made it possible to group the investigations by practical approaches. Seven groups of methods were categorized, namely:

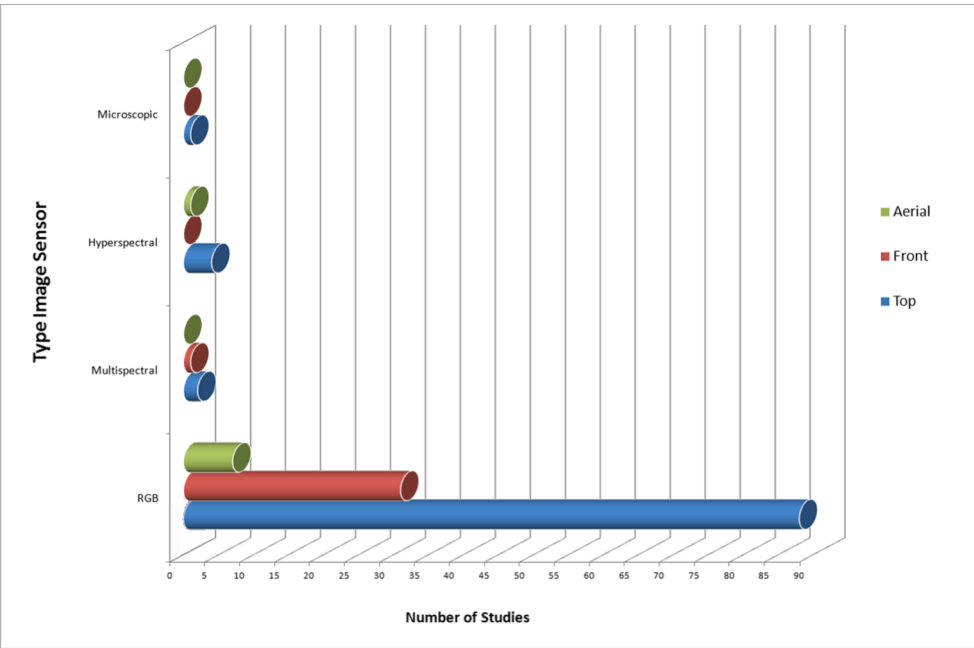


Fig. 6. Image characteristics according to the type of sensor and perspective of acquisition.

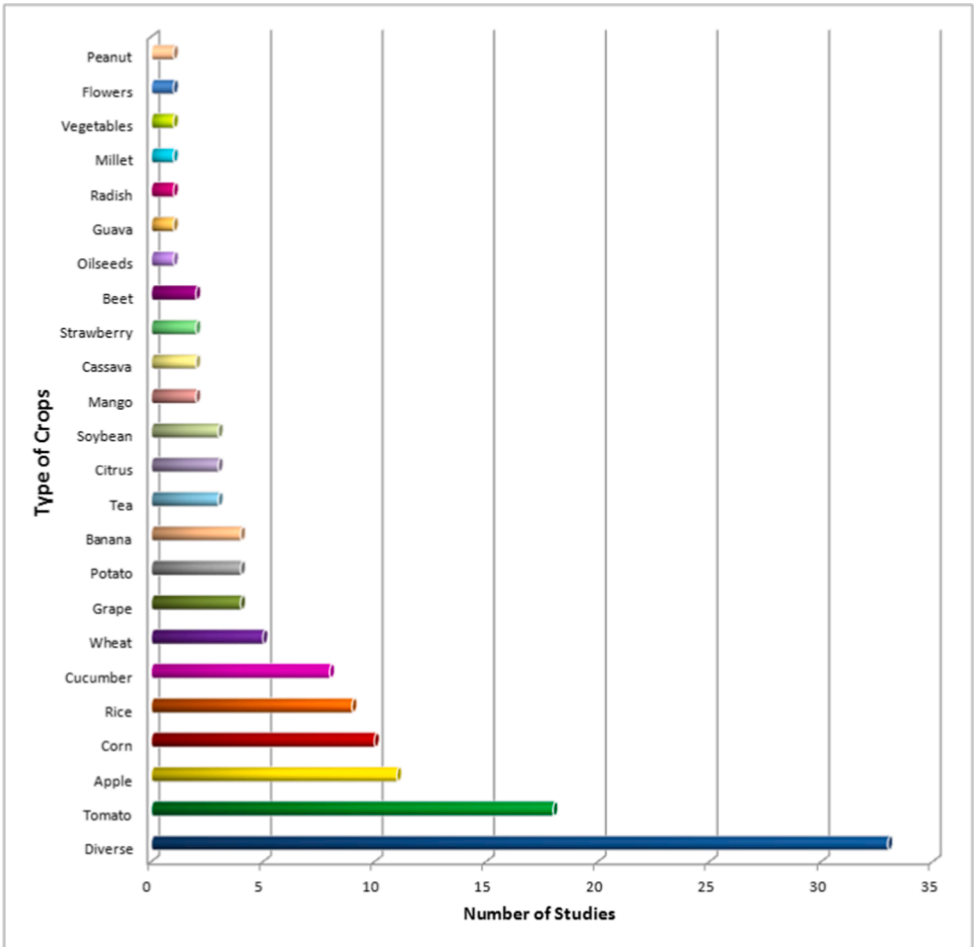


Fig. 7. The most investigated crops in the 121 studies that apply CNN's to identify diseases.

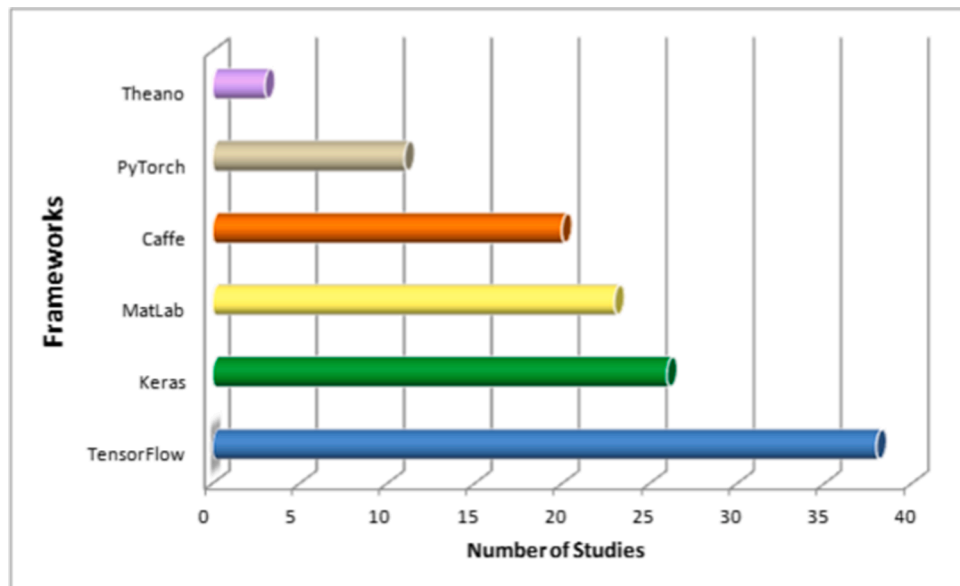


Fig. 8. The most used frameworks among the 121 studies summarized.

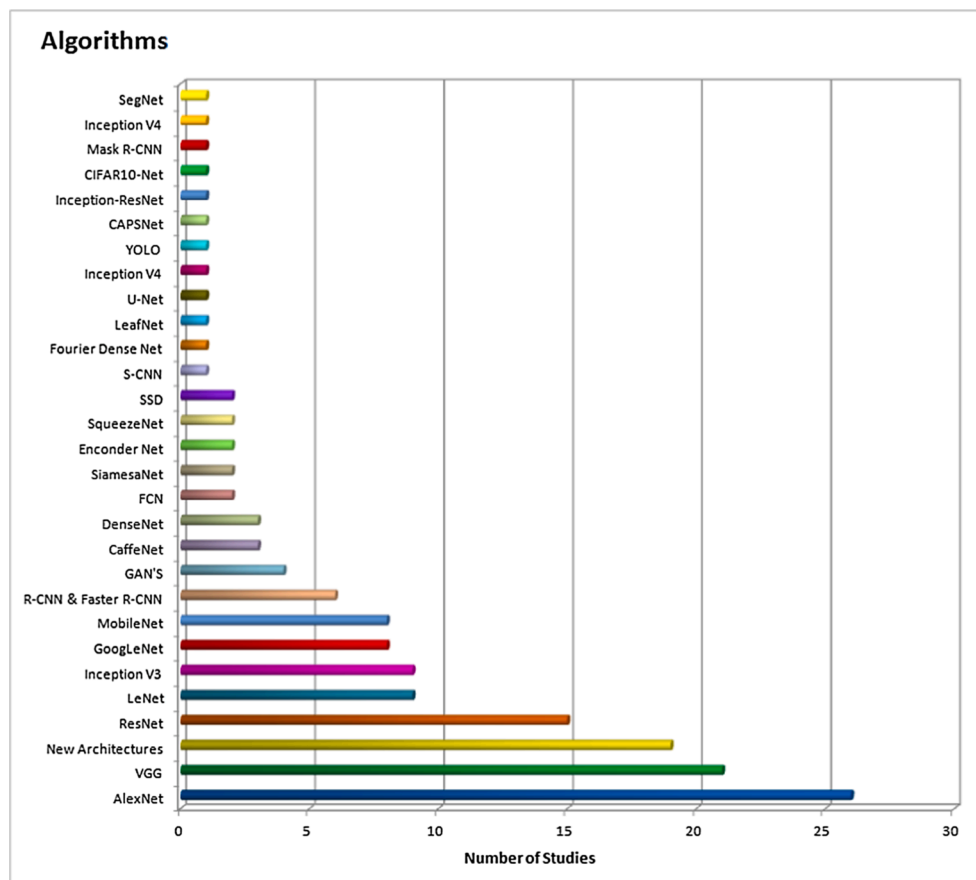


Fig. 9. The CNN Algorithms most used in the 121 summarized studies.

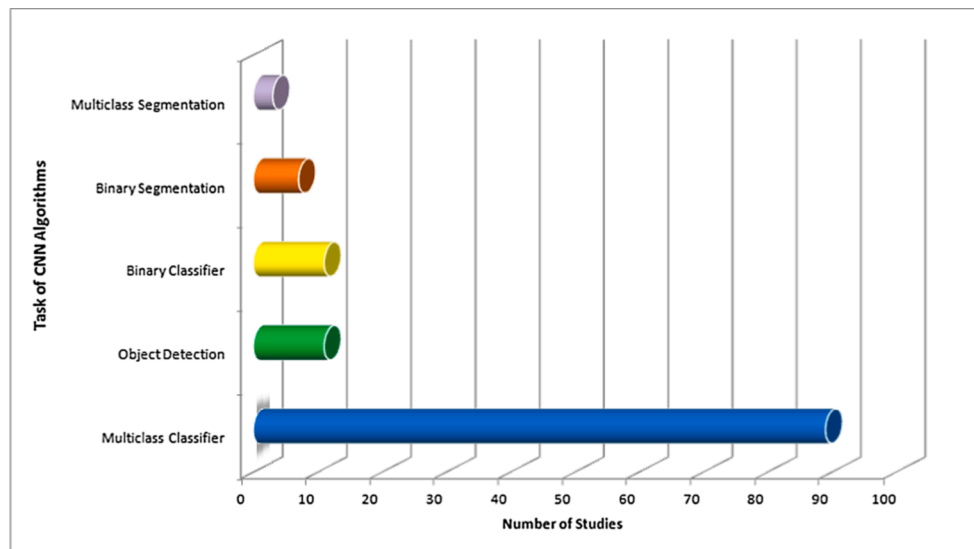


Fig. 10. Tasks of CNN Algorithms categorized according to the summary of information from the 121 studies.

Customizing Layers, Transfer Learning, New Architecture, Fine Tune and hyperparameters, Image Segmentation, Data Augmentation, and Unsupervised Learning.

Among the 121 studies evaluated, it is possible to notice that only 25 studies (S1, S2, S6, S7, S11, S12, S13, S15, S22, S35, S43, S54, S61, S65, S68, S70, S75, S84, S86, S91, S100, S113, S114, S116, S120) propose a new architecture. In Fig. 4, the distribution of studies by practical approaches is presented.

4.2. SQ2: What are the characteristics of the datasets predominantly used?

Regarding the predominance and characteristics of the datasets most used in the investigated studies, we highlight the customized initiatives designed mainly for a research need and the PlantVillage (Hughes and Salathé, 2016) dataset. Fig. 5 shows the distribution of studies by dataset and type of image capture environment.

The customized datasets are present in 66 studies (S1, S2, S3, S6, S7, S8, S9, S12, S15, S17, S18, S20, S22, S24, S25, S28, S32, S33, S34, S36, S37, S38, S40, S43, S44, S45, S46, S47, S49, S50, S52, S58, S64, S68, S69, S70, S71, S73, S74, S75, S77, S79, S80, S82, S84, S87, S88, S89, S90, S91, S92, S95, S96, S97, S98, S100, S102, S103, S104, S107, S112, S113, S115, S116, S119, S121) and PlantVillage is used in your full or partial version in 45 studies (S3, S4, S9, S11, S13, S14, S18, S21, S23, S26, S27, S28, S29, S35, S41, S48, S51, S53, S55, S56, S57, S60, S61, S62, S63, S65, S66, S67, S68, S69, S72, S73, S76, S78, S81, S85, S86, S93, S99, S101, S105, S108, S110, S114, S117).

The diversity of sensors and the different perspectives on capturing images are also decisive factors for the characterization of the data set. In summarizing the studies, four types of image categories were identified: RGB, Multispectral, Hyperspectral, and Microscopic. Including three perspectives for image acquisition: Top View, Front View, and Aerial View. We summarize the information extracted from all 121 studies from the final selection, presented in Fig. 6.

Many studies used a dataset of RGB images in Top View, totaling 88 studies (S1, S2, S3, S4, S5, S9, S11, S13, S14, S16, S17, S18, S19, S21, S23, S26, S27, S28, S29, S31, S32, S34, S35, S36, S38, S39, S40, S41, S42, S45, S46, S48, S51, S53, S55, S56, S57, S58, S59, S60, S61, S62, S63, S66, S67, S68, S69, S72, S73, S76, S78, S79, S81, S84, S86, S89, S93, S94, S96, S99, S101, S105, S107, S108, S109, S110, S111, S113, S114,

S116, S118, S119, S120, S121). Studies based on RGB images in Frontal View, a total of 31 studies (S6, S7, S10, S12, S15, S20, S24, S30, S43, S44, S49, S50, S52, S64, S65, S71, S75, S77, S83, S87, S90, S91, S92, S98, S100, S102, S103, S104, S106, S115, S117) and RGB images with Aerial View, only 7 studies (S8, S47, S54, S82, S95, S97, S112).

The 121 summarized studies, three studies used multispectral images, being the studies S25 and S33 applying the Top View perspective and in S85 a Frontal View perspective. Five studies used hyperspectral images, four (S37, S70, S74, S80) in the Top View perspective, and the S22 study with an aerial view perspective. Only the S88 study uses microscopic images in its data set.

4.3. SQ3: What types of crops are most investigated with approaches using CNN's?

The types of crops present in the approaches that use CNN's to identify diseases are summarized graphically and presented in Fig. 7. It is possible to observe that the type called "Diverse" categorizes all studies that use more than one type of crop in their approaches and consequently make use of a database with diversified plant phenotypes. In particular, this is a predominant feature in the PlantVillage dataset.

Studies that investigate only one type of crop in particular highlight those that address the phytopathology of Tomato crop with 18 investigations (S3, S9, S27, S44, S48, S55, S58, S61, S67, S70, S75, S78, S86, S93, S99, S102, S105, S108).

4.4. SQ4: What types of approaches and frameworks are commonly used?

In the summary process, it was found that the investigated studies make use of 6 main Frameworks, used to implement solutions involving machine learning, as well as predictive analysis. Fig. 8 is presented the distribution of studies grouped by the framework used in the approach development process.

It was possible to verify among the 121 summarized studies, the TensorFlow was the most used framework in plant diseases using convolutional neural networks. A total of 38 investigations (S1, S2, S3, S5, S8, S11, S12, S14, S15, S23, S25, S27, S28, S38, S39, S40, S42, S44, S50, S53, S55, S56, S61, S62, S63, S64, S65, S66, S68, S69, S70, S76, S77, S81, S86, S104, S106, S110) used the TensorFlow.

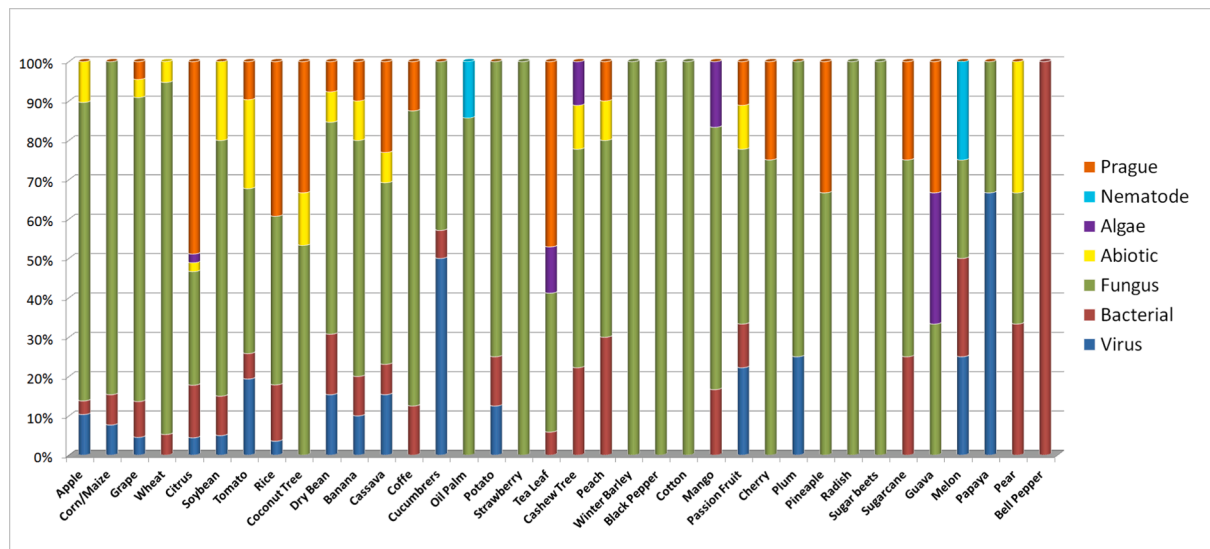


Fig. 11. Relationship of type of crops versus groups of pathogens versus occurrences of diseases investigated by causative agents.

4.5. SQ5: Which CNN algorithm is prevalent in current approaches?

Deep learning is not a single approach, but a class of algorithms and topologies that can be applied to a wide variety of problems. In Fig. 9, it is possible to follow the distribution of these algorithms grouped according to the 121 summarized studies.

Thirty algorithms used in the approaches were accounted for, with predominant algorithms that define the classic architectures, with emphasis on AlexNet and VGG. It should be noted that 19 studies (S1, S6, S7, S11, S12, S13, S22, S35, S43, S61, S68, S70, S75, S84, S86, S91, S100, S116, S120) propose new architectures and 15 (S9, S18, S19, S23, S32, S34, S39, S41, S42, S49, S50, S51, S53, S65, S103) investigations customize classic algorithms to solve the problem of identifying plant diseases using CNN's.

Additionally, we analyze what all the CNN algorithms perform (i.e., classification tasks) in the 121 summarized studies and categorize them into five types focused on the primary tasks: Multiclass Classifier, Binary Classifier, Multiclass Segmentation, Binary Segmentation and Object Detection. In Fig. 10, we present the categorization of these studies according to each type of task performed by CNN.

We found the prevalence of CNN algorithms that perform a multiclass classification task (several types or stages of plant disease) in 89 studies (S1, S4, S6, S9, S11, S12, S13, S14, S16, S18, S19, S21, S23, S26, S27, S28, S29, S30, S31, S32, S34, S38, S39, S41, S42, S43, S46, S48, S50, S51, S52, S53, S54, S55, S56, S57, S58, S59, S60, S61, S62, S63, S64, S65, S67, S68, S69, S71, S72, S73, S74, S76, S77, S78, S79, S81, S82, S83, S84, S85, S86, S87, S89, S90, S91, S92, S93, S97, S98, S99, S100, S101, S104, S105, S106, S108, S109, S110, S111, S112, S113, S114, S115, S116, S117, S118, S119, S120, S121). However, the Binary Classification task (diseased or healthy) raises in only 11 studies (S7, S10, S22, S36, S37, S47, S49, S66, S88, S95, S103).

For the Object Detection task focused on to identify the position of the diseased plant (or the part affected by the disease or by an agent causing the plant disease), only 11 studies were identified (S2, S15, S20, S24, S35, S40, S44, S45, S75, S94, S102).

Seven Approaches (S8, S17, S25, S33, S70, S80, S107) implemented a Binary Segmentation task, dividing the image into homogeneous regions according to a defined criterion, generating binary images of the plant diseases. Finally, three studies were found (S3, S5, S96) that implement the Multiclass Classification task (or pixel labeling)

addressed to label every pixel in an image.

4.6. SQ6: What types of plant diseases (biotic causative agents) are most investigated with approaches using CNN's?

Using the data summarized in Appendix B, we analyzed 54 datasets with 142 types of crops, associating each disease-causing pathogen to its large group, namely: Virus, Bacteria, Fungi, Algae, Plague, Nematodes and, Abiotics. In Fig. 11, we present the type relationship of the crops with groups of pathogens, quantifying the number of occurrences of diseases investigated by their causative agents.

5. Discussion

To keep the discussion in line with the core questions, all of them answered in the previous section, our study is organized into four topics that describe state of the art on identifying plant diseases using CNN's.

5.1. Predominant approaches

Different innovations in CNN architectures have been proposed since 1998 with the presentation of LeNet-5 (Lecun et al., 1998) and can be categorized as parameter optimization, regularization, structural reformulation (Convolutional layer, Polling layer, and Activation Function), loss function and fast processing. However, it is observed that the main improvement in the performance of CNN's is motivated by the structural reformulation and design of new blocks (Khan et al., 2019).

When we tried to answer the question SQ1 of Section 4.1, we noticed that the investigations that applied state of the art for identification and classification of plant diseases using CNN's explored different innovative approaches to provide a reliable solution to the problem investigated.

As for the new architectures, we noticed three trends for the formulation of approaches. One of them is the structural reformulation of traditional architectures, such as LeNet (studies S61, S65, S68, S84, S100, S114, and S116), AlexNet (S113 and S120) and ResNet (S6, S43, S75, and S86). Another trend is the blending of blocks between architectures, such as the S15 study (Yolo-DenseNet) and the S22 study (Inception-ResNet). The latest trend includes proposals that implement architectures explicitly designed for the identification and classification

of plant diseases, with emphasis on models that use hyperspectral images, namely: GPDCNN (S1), 3D-CNN (S7), PDDNN (S11), M-bCNN-CKM (S12), Teacher/ Student Architecture (S13), PlantDiseaseNet (S35), Fourier Dense Network (S54) and OR-AC-GAN (S70).

Another relevant factor regarding the predominant approaches is the intrinsic relationship between the three highlighted categories (Customizing layers, New Architecture, Fine-tune, and Hyperparameters). The investigated studies include categories sharing the same challenges, such as, for example, reducing the number of model parameters, reducing the size of the model, the need to optimize hyperparameters, and an improvement in the prediction capacity adjusted at a cost acceptable computational.

When analyzing the categories with customization of layers and new architectures, it is evident that when we deal with the challenges related to the identification of plant diseases, the generated models take a reverse path of traditional architectures. While traditional architectures, such as, for example, AlexNet, VGG16, and ResNet built their notoriety for their ability to generalize, the models investigated tend to be increasingly specialized in a specific type of crops. This phenomenon can be explained, considering that the architectures and models used to identify plant diseases and, consequently, the datasets used, can only establish a pattern for the extraction of discriminating features using the plant's morphological characteristics, considering that a given pathogen can cause different symptoms and injuries depending on the type of crops evaluated.

This specialization feature of the approaches that implement layer customization and new architectures, although not mentioned in any of the summarized studies, refers to a most popular topic in machine learning in recent years (e.g., Neural Architecture Search (NAS)), with many studies focusing on search strategies as reinforcement learning, evolutionary algorithms, and Bayesian optimization (Elsken et al., 2019).

Most research using NAS shows a bias inference, such as, for example, the high computational cost to establish a CNN architecture automatically; the need for prior knowledge about the features of the dataset, which means that it is not suitable for new domains; and finally, the prioritization of inference performance, which tends to produce a large architecture (Jin et al., 2019). Despite these challenges, a significant absence of this approach was noted in investigations to identify plant diseases using CNN's.

5.2. Characteristic of the datasets

The characteristics of the datasets, such as the number of samples, number of diseases, symptomatic similarity caused by pathogens, pests, and abiotic factors, constitute one of the most important limitations regarding the accuracy of the approaches that implement the identification and classification of diseases of plants. Experimental results indicate that while the technical constraints linked to automatic plant disease classification have mainly been overcome, the use of limited image datasets for training brings many undesirable consequences that still prevent the effective dissemination of this type of technology (Barbedo, 2018).

The answer to question SQ2 from Section 4.2, groups the datasets with the highest recurrence in the studies, and demonstrates that the approaches try to reduce the limitations by creating customized sets that mix images from controlled environments with images captured under real growing conditions. However, we noticed that 65.28% of the approaches use sets with images in controlled environments, with PlantVillage being present in 37.19% of the summarized studies.

It should be noted that many approaches, such as studies S28, S35, S40, S65, S69, and S73, have already demonstrated a reduction in accuracy, when they use models trained exclusively with sets of images

acquired in controlled environments to classify images in challenging conditions such as illumination, complex background, different resolution, size, pose, and orientation of real scene images. According to Goncharov et al. (2019), who proposed the approach of the S28 study, when applying a customized CNN architecture, they obtained high precision with training and testing using the PlantVillage dataset. However, the results obtained when applying the same network in images acquired in real crop conditions, there was a reduction of about 30% to 40% in the global accuracy.

Additionally, we categorized the studies according to the type of sensor used in the capture step (RGB, Multispectral, Hyperspectral, and Microscopic) and the image acquisition perspective (Aerial View, Front View, and Top View) that compose the investigated datasets. RGB images with a top view are predominant, considering two factors: i) the wide use of the PlantVillage dataset by the proposed approaches, composed of RGB images with a top view in controlled environments and; ii) the use of low-cost RGB sensors combined with ease image handling and the representative efficiency of the images.

Studies using RGB images with Frontal View show significant advances, given the popularization of smartphones to capture images from plant diseases in real cultivation environments. However, it should be noted that even with the representative efficiency of RGB images, the frontal perspective raises problems as distortion, affecting the geometry of the captured images.

The use of aerial images for diagnosis and monitoring of crop areas is a trend in agriculture, mainly caused by advances in unmanned aerial vehicles (UAV), allowing them to obtain images of large areas at low cost and high resolution. Despite the studies S22, S54, S82, S97, and S112 demonstrate robust solutions using datasets with an area perspective, this type of image has a peculiarity that can compromise the performance of CNN's. Ground Sample Distance (GSD) is a factor already known in photogrammetry surveys, so there is a relationship between the height of flight and the type of camera used that will cause a variation in the GSD, directly implying the representation quality of the pixels of the captured image.

It was not possible to find in the evaluated studies any dataset that included images of crops using covered large areas of cultivation. Specifically, in this case, the economically relevant crops, such as rice, soybeans, wheat, corn, and cotton. Nor did we find an approach that used a dataset from these crops that could go beyond binary classification (diseased or healthy) in the process of identifying plant diseases using CNN's. The process of annotating these datasets, for example, is very laborious, which can explain this absence.

Another critical factor is the promising use of Multispectral and Hyperspectral images in identifying plant diseases using CNN's. The studies S22, S37, S70, S74, and S80 demonstrate relevant results, even though the vast majority of experiments are conducted with datasets formed by images captured in controlled environments. Additionally, it should be noted that the cost of Multispectral and Hyperspectral sensors for image acquisition is still high, and the complexity of preprocessing and annotation of these types of images is inherent to the number of the used spectral bands.

Most of the investigated studies use a visual diagnosis of the plant, which consists of comparing the aspect of a plant with a problem against a healthy pattern one. However, the correct classification of pathogens that cause plant diseases is based on laboratory analysis using microscopy. Only the S88 study uses microscopic images to classify disease-causing pathogens, even though it has a low inference capacity.

5.3. CNN algorithms and frameworks

When evaluating the most used Frameworks by the selected studies, and the predominance of TensorFlow (Abadi et al., 2015) is evident,

followed by the API Keras (Chollet et al., 2015) with TensorFlow in the back-end. However, it is not explicit in the evaluated studies a possibility to suggest Frameworks with extensive documentation, application examples, tutorials, usability, scalability and a developer community predominant in the presented approaches.

When implementing these CNN's architectures, we note in traditional algorithms, for example, AlexNet (Krizhevsky et al., 2012), VGG (Simonyan and Zisserman, 2014), ResNet (He et al., 2016), LeNet (Lecun et al., 1998), InceptionV3 (Szegedy et al., 2015b), GoogLeNet (Szegedy et al., 2015a), are predominant in the selected studies. The recurring use of these implementations is explained by their success cases based on the ImageNet challenge (Russakovsky et al., 2014). Approximately 85% of the approaches use the transfer learning, fine-tune, and hyper-parameters methods to increase the accuracy of their results in detriment to the deficiencies of the used datasets.

The work of Krizhevsky et al. (2012) that presents the AlexNet architecture is one of the most cited and influential works on computer vision applied for object classification based on convolutional neural networks. When observing the prevalence and the influence of this architecture among the summarized studies focused on identifying plant diseases, it is evident that this model is changed and combined in an easy implementation and almost all of them using low computational cost. Among all the approaches that use layers customization of the architectures or propose new models, AlexNet is the basis for most investigated solutions. It allows the researchers addressed to conceptual flexibility in this kind of architecture, prospects new solutions using high inference capacity and low computational complexity.

It should also be noted that among the tasks implemented by the CNN's algorithms, the classification of the types of diseases is prevalent. We note that there is a growing trend of solutions that implement the task of object detection, intuiting the classification and the correct position of the areas affected by the disease in the plant. Additionally, we note that among the investigated studies there is no approach that implements a CNN algorithm for object detection using a dataset with aerial images, capable of classifying multiple diseases and determining the position of the affected plants in the cultivation areas. Bearing in mind that there are many times in real cultivation environments the simultaneous action of several pathogens that cause plant diseases.

5.4. Types crops and disease causing pathogens

The results showed that in addition to the approaches that investigate a diverse set of crops, tomatoes together with apples, corn, rice, cucumbers, wheat, grapes, potatoes, and bananas are the ones that concentrate the most significant number of investigations using CNN's. Taking as a concept that grains and cereals are the primary sources of energy ingested by humans, it was evident the lack of studies with approaches that primarily contemplated these crops.

In the case of soybeans, one of the main legumes of industrial interest, it was possible to identify only studies S7, S81, and S84. Highlighted is the approach proposed by Nagasubramanian et al. (2019) in study S7 that creates an architecture called 3D-CNN that can be used to extract features jointly across the spatial and spectral dimension for classification of a 3D hyperspectral data.

When analyzing the results that deal with the types of plant diseases, we establish a relationship between the diseases investigated in the approaches with their causative agents. It was possible to observe that the vast majority of investigations identify and classify diseases caused by fungi. This fact can be explained because it is estimated that 70% of the main plant diseases are caused by this pathogen (Agrios, 2005;

Godfray et al., 2016).

The diseases caused by viruses and bacteria, although they do not have consistent quantitative approaches, are represented in a diversity of crops found in the datasets. The diseases caused by algae are represented by the crop Cashew, Citrus, Guava, Strawberries, and tea leaves. We also identified the classification of diseases caused by abiotic factors in investigations using CNN's.

The inexpressive number of approaches that classify diseases caused by nematodes caught our attention. These pathogens are considered to be one of the most significant phytosanitary risks in the main grain and cereal crops in the world. Indeed, the absence of investigations using CNN's to identify symptoms caused by nematodes is the symptomatic similarity of the characteristics presented by the plant with other diseases or particularities that are not visibly expressed in the leaves and stems of the plant.

6. Conclusions

The automatized plant disease detection systems propose the identification and classification of plant diseases, reconciling the know-how of specialists in phytopathology and the ability to extract symptomatic features, through convolutional neural networks.

The diversity of problems and the specificities of real-world scenarios increases the difficulty to semantically catalog the data in representative data sets with a sufficient number of labeled samples. This problem becomes a relevant and challenging bottleneck to develop machine learning methods more applicable in practice.

Our results demonstrate the significant advances in the use of CNN in plant disease prediction processes. It was possible to observe that the traditional architectures combined with optimization and customization methods, despite the complexity of the dataset composed with images captured in real environments of the crops, present relevant accuracy. The tendency of many approaches that propose new CNN architectures based on the process of identification of plant diseases is growing. However, a specialization of these architectures was observed to the detriment to the type of crop investigated.

Although the symptomatic diversity caused by the same pathogen is a complicating factor to the inference of plant disease detection and recognition, it was noted a lack of studies that propose CNN architectures that aim to specifically disease regardless of the type of crops.

Our investigation shows that there is an increasing trend in approaches combining CNN's with multispectral and hyperspectral images with more information potential than RGB images. We also highlight the use of high-resolution images captured from UAV to identify plant diseases as an emerging and promising trend with numerous research challenges available. Finally, we also would want to warn about grains and cereals with high food and financial representativeness, which were often overlooked by almost all the approaches. Also, we highlight the inexpressive number of methods that identify or classify diseases caused by nematodes in all of them developed techniques.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Summarization

Table A.3.

Table A.3

Characteristics of the Primary Studies of the Final Selection.

ID	Refs	Year	IQ	Algorithm	Task of Model	Best Accuracy	Dataset Characteristics	Type of Crops	Type of Images	View Point of Images	Language & Framework	Predominant Approach
S1	Zhang et al. (2019)	2019	5	GPDCNN	Multiclass Classifier	94,65%	Customized 700 Images/ 7 Class Real Environment	Cucumbers	RGB	Top View	TensorFlow	New Architecture
S2	Hasan et al. (2019)	2019	4,5	R-CNN & Faster R-CNN	Object Detection	93,4%	Customized 335 Images/ 3 Class Real Environment	Wheat	RGB	Top View	TensorFlow	Transfer Learning
S3	Sharma et al. (2019)	2019	4,5	S-CNN	Multiclass Segmentation	98,6%	PlantVillage Subset 15.817 Images/ 8 Class Controlled Environment	Tomato	RGB	Top View	TensorFlow	Image Segmentation
S4	Liang et al. (2019)	2019	4,5	ResNet	Multiclass Classifier	98%	Customized 17.000 Images/ 10 Class Real Environment	Diverse	RGB	Top View	PyTorch	Fine Tune & Hyperparameters
S5	Huang et al. (2019)	2019	4,5	UNet + InceptionV3	Multiclass Segmentation	87,45%	PlantVillage 32.933 Images/ 45 Class Controlled Environment	Diverse	RGB	Top View	TensorFlow	Image Segmentation
S6	Picon et al. (2019)	2019	4	RESNET-MC1	Multiclass Classifier	98% BAC	Crop Disease AI Challenger 2018 40.000 Images/ 19 Class Controlled Environment	Diverse	RGB	Front View	TensorFlow/Keras	New Architecture
S7	Nagasubramanian et al. (2019)	2019	4	3D-CNN	Binary Classifier	95,73%	Customized 1.090 Images/ 2 Class Controlled Environment	Soybean	Hyperspectral	Front View	TensorFlow/Keras	New Architecture
S8	Wiesner-Hanks et al. ((2019) 1550.)	2019	4	ResNet - Crowdsourced	Binary Segmentation	99,79%	Customized 15.240 Images/ 2 Class Real Environment	Corn	RGB	Aerial View	TensorFlow	Image Segmentation
S9	Zhang et al. (2019)	2019	4	Three Channel CNN AlexNet	Multiclass Classifier	91,15% 94,27%	PlantVillage Subset 15.817 Images/ 8 Class Controlled Environment	Tomato	RGB	Top View	MatLab	Layers Custom
S10	Liang et al. (2019)	2019	4	LeNet	Binary Classifier	95,83%	Customized 500 Images/ 5 Class Real Environment	Cucumbers				
S11	Hari et al. (2019)	2019	4	PDDNN	Multiclass Classifier	86%	Rice Blast Disease 5.808 Images/ 2 Class Controlled Environment	Rice	RGB	Patches Front View	PyTorch	Layers Custom
S12	Lin et al. (2019)	2019	4	M-bCNN-CKM	Multiclass Classifier	96,5%	PlantVillage 14.810 Images/ 10 Class Controlled Environment	Diverse	RGB	Top View	TensorFlow	New Architecture
S13	Brahimi et al. (2019)	2019	4	AlexNet VGG Teacher/Student Architecture	Multiclass Classifier	83,1% 88,5% 90,7% AOPC	Customized 16.652 Images/ 8 Class Real Environment	Wheat	RGB	Front View	TensorFlow	New Architecture
S14	Khandelwal and Raman (2019)	2019	4	ResNet	Multiclass Classifier	99,37%	PlantVillage 54.306 Images/ 38 Class Controlled Environment	Diverse	RGB	Top View	TensorFlow/Keras	New Architecture
							PlantVillage 86.198 Images/ 57 Class Controlled Environment	Diverse	RGB	Top View	TensorFlow	Transfer Learning

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Table A.3 (continued)

ID	Refs	Year	IQ	Algorithm	Task of Model	Best Accuracy	Dataset Characteristics	Type of Crops	Type of Images	View Point of Images	Language & Framework	Predominant Approach
S15	Tian et al. (2019)	2019	4	YOLOV3 - DenseNet	Object Detection	95,75%	Customized 640 Images/ 2 Class Real Environment	Apple	RGB	Front View	TensorFlow	New Architecture
S16	Barbedo (2019)	2019	4	GoogLeNet	Multiclass Classifier	94%	PDDb Embrapa 46.409 Images/ 79 Class Real Environment	Diverse	RGB	Top View	MatLab	Transfer Learning
S17	Abdalla et al. (2019)	2019	4	VGG16 Encoder	Binary Segmentation	96%	Customized 400 Images/ 2 Class Controlled Environment	Oilseeds	RGB	Top View	MatLab	Fine Tune & Image Segmentation
S18	Singh et al. (2019)	2019	4	MCNN - AlexNet	Multiclass Classifier	97,13%	PlantVillage Subset (1130) Custom Mango Leaf (1070) 2.200 Images/ 2 Class Controlled Environment e Real	Mango	RGB	Top View	TensorFlow/Keras	Layers Custom
S19	Long and Liu (2019)	2019	4	ResNet-B VGG16 InceptionV3 MobileNet	Multiclass Classifier	99,7% 94,7% 99,5% 99,6%	AI Challenger 40.772 Images/ 23 Class Controlled Environment	Diverse	RGB	Top View	PyTorch	Layers Custom
S20	Selvaraj et al. (2019)	2019	4	Faster R-CNN InceptionV2 Faster R-CNN ResNet50 SSD MobileNetV1	Object Detection	99,0% 99,0% 98,2%	Customized 18.000 Images/ 2 Class Real Environment	Banana	RGB	Front View	TensorFlow/Keras	Layers Custom
S21	Kc et al. (2019)	2019	3,5	MobileNet	Multiclass Classifier	97,65%	PlantVillage 54.306 Images/ 38 Class Controlled Environment	Diverse	RGB	Top View	TensorFlow/Keras	Layers Custom
S22	Zhang et al. (2019)	2019	3,5	Reduce MobileNet DCNN	Binary Classifier	98,34% 85,0%	Customized 15.000 Images/ 2 Class Controlled Environment	Wheat	Hyperspectral	Aerial View	PyTorch	New Architecture
S23	Alruwaili et al. (2019)	2019	3,5	Inception-ResNet Block DCNN based AlexNet	Multiclass Classifier	99,11%	PlantVillage 54.306 Images/ 38 Class Controlled Environment	Diverse	RGB	Top View	TensorFlow	Layers Custom
S24	Rancon et al. (2018)	2019	3,5	MobileNet	Object Detection	91,0%	Customized 6.000 Images/ 2 Class Real Environment	Grape	RGB	Front View	TensorFlow/Keras	Transfer Learning
S25	Douarre et al. (2019)	2019	3,5	SegNet e GANs	Binary Segmentation	≈ 64,3%	Customized 50 Images/ 2 Class Real Environment	Apple	Multispectral	Top View	TensorFlow	Data Augmentation
S26	Aravind et al. (2019)	2019	3,5	AlexNet	Multiclass Classifier	99,16%	PlantVillage Subset 4063 Images/ 4 Class Controlled Environment	Grape	RGB	Top View	MatLab	Layers Custom
S27	Khamparia et al. (2019)	2019	3,5	Convolutional Enconder Network	Multiclass Classifier	99,9%	PlantVillage Subset 900 Images/ 6 Class Controlled Environment	Corn Tomato Potato	RGB	Top View	Python	Layers Custom & Hyperparameters
S28	Goncharov et al. (2019)	2019	3,5	ResNet SiameseNet	Multiclass Classifier	99,4% 94,3%	PlantVillage Subset 4063 Images/ 4 Class Controlled Environment	Grape	RGB	Top View	TensorFlow	Layers Custom
				ResNet SiameseNet		98,0% 92,0%						

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Table A.3 (continued)

ID	Refs	Year	IQ	Algorithm	Task of Model	Best Accuracy	Dataset Characteristics	Type of Crops	Type of Images	View Point of Images	Language & Framework	Predominant Approach
							Customized Internet 170 Images/ 4 Class Real Environment					
S29	Toda and Okura (2019 (2019))	2019	3,5	InceptionV3	Multiclass Classifier	97,14%	PlantVillage 54.306 Images/ 38 Class Controlled Environment	Diverse	RGB	Top View	TensorFlow/Keras	Layers Custom 75% reduction in parameters
S30	Türkoglu and Hanbay (2019)	2019	3,5	ResNet	Multiclass Classifier	97,86%	Turquia Customized 1.965 Images/ 8 Class Controlled Environment	Diverse	RGB	Front View	MatLab	Layers Custom
S31	Howlader et al. (2019)	2019	3,5	AlexNet	Multiclass Classifier	98,74%	BU-Guava-Leaf 2.075 Images/ 4 Class Controlled Environment	Guava	RGB	Top View	TensorFlow/Keras	Fine Tune & Hyperparameters
S32	Jiang et al. (2019)	2019	3,5	VGG-INCEP	Multiclass Classifier	97,14%	ALDD 26.377 Images/ 6 Class Real Environment	Apple	RGB	Top View	Caffe	Layers Custom
S33	Lin et al. (2019)	2019	3,5	U-Net Segmentation	Binary Segmentation	96,08%	Customized 50 Images/ 2 Class Controlled Environment	Cucumbers	Multispectral	Top View	TensorFlow/Keras	Data Augmentation & Image Segmentation
S34	Labaña et al. (2019)	2019	3,5	Baseado LeNet	Multiclass Classifier	91,0%	Customized 2000 Images/ 4 Class Controlled Environment	Citrus	RGB	Top View	TensorFlow/Keras	Layers Custom & Data Augmentation
S35	Arsenovic et al. (2019)	2019	3,5	PlantDiseaseNet Architecture	Object Detection	93,67%	PlantVillage (54.306) PlantDataSet (24.959) 79.265 Images/ 42 Class Controlled Environment	Diverse	RGB	Top View	TensorFlow/Keras	New Architecture
S36	Coulibaly et al. (2019)	2019	3,5	VGG16	Binary Classifier	95,0%	Customized 124 Images/ 2 Class Controlled Environment	Millet	RGB	Top View	TensorFlow/Keras	Transfer Learning
S37	Polder et al. (2019)	2019	3,5	FCN	Binary Classifier	88,0%	Customized 22.000 Images/ 2 Class Controlled Environment	Potato	Hyperspectral	Top View	PyTorch	Fine Tune & Hyperparameters & Image Segmentation
S38	Hu et al. (2019)	2019	3,5	CIFAR-10Net	Multiclass Classifier	95,5%	Customized 144 Images/ 4 Class Real Environment	Tear Leaf	RGB	Top View	TensorFlow	Layers Custom
S39	Yu and Son (2019)	2019	3,5	ROI-aware DCNN Based VGG	Multiclass Classifier	84,3%	Apple Research Soul Korea 865 Images/ 4 Class Real Environment	Apple	RGB	Top View	TensorFlow	Transfer Learning & Image Segmentation
S40	Ramcharan et al. (2019)	2019	3,5	MobileNet & SSD	Object Detection	94,0%	IITA Cassava 2.017 Images/ 7 Class Controlled Environment	Cassava	RGB	Top View	TensorFlow	Transfer Learning
						75,0%	Customized 1.375 Images/ 7 Class Real Environment					
S41	G.G. and A.P.J. (2019)	2019	3,5	Custom CNN Based AlexNet	Multiclass Classifier	96,46%	PlantVillage 54.306 Images/ 38 Class Controlled Environment	Diverse	RGB	Top View	TensorFlow/Keras	Layers Custom
S42	Hang et al. (2019)	2019	3	Custom Inception	Multiclass Classifier	91,7%	IA Challenger Plant Leaf Disease 4.523 Images/ 10 Class Controlled Environment	Diverse	RGB	Top View	TensorFlow	Layers Custom

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Table A.3 (continued)

ID	Refs	Year	IQ	Algorithm	Task of Model	Best Accuracy	Dataset Characteristics	Type of Crops	Type of Images	View Point of Images	Language & Framework	Predominant Approach
S43	Sibiya and Sumbwanyambe (2019)	2019	3	CNN 50 hidden layers	Multiclass Classifier	92,85%	Customized 100 Images/ 3 Class Real Environment	Corn	RGB	Front View	MatLab	New Architecture
S44	Wang et al. (2019)	2019	3	Mask R-CNN e ResNet101	Object Detection	99,64%	Customized 1.430 Images/ 11 Class Real Environment	Tomato	RGB	Front View	TensorFlow	Image Segmentation
S45	Ozguven and Adem (2019)	2019	3	Faster R-CNN	Object Detection	95,48%	Customized 155 Images/ 4 Class Real Environment	Sugar-beet	RGB	Top View	MatLab	Layers Custom
S46	Chen et al. (2019)	2019	3	LeafNet	Multiclass Classifier	99,84%	Customized 7.905 Images/ 7 Class Controlled Environment	Tear Leaf	RGB	Top View	MatLab	Fine Tune & Hyperparameters
S47	Wu et al. (2019)	2019	3	ResNet-34	Binary Classifier	95,1%	Customized 6.267 Images/ 2 classes Real Environment	Corn	RGB	Aerial View	PyTorch	Fine Tune & Hyperparameters
S48	Ashqar and Abu-Naser (2019)	2019	3	LeNet	Multiclass Classifier	99,84%	PlantVillage Subset 9.000 Images/ 6 Class Controlled Environment	Tomato	RGB	Top View	Caffe	Layers Custom
S49	Zhang et al. (2019)	2019	3	DCGAN + Custom AlexNet	Binary Classifier	90,0%	Customized 800 Images/ 2 Class Real Environment	Citrus	RGB	Front View	TensorFlow/Keras	Layers Custom e Data Augmentation
S50	Xing et al. (2019)	2019	3	WeaklyDenseNet-16	Multiclass Classifier	99,83%	Customized 3.510 Images/ 17 Class Controlled Environment	Citrus	RGB	Front View	TensorFlow	Layers Custom
S51	Abade et al. (2019)	2019	3	M-AlexNet	Multiclass Classifier	99,59%	PantVillage 54.306 Images/ 38 Class Controlled Environment	Diverse	RGB	Top View	TensorFlow/Keras	Fine Tune & Hyperparameters
S52	Dawei et al. (2019)	2019	3	M-GoogLeNet AlexNet	Multiclass Classifier	99,55% 98,92%	Customized 484 Images/ 10 Class Real Environment	Tear Leaf	RGB	Front View	MatLab	Transfer Learning
S53	Kaya et al. (2019)	2019	3	VGG16	Multiclass Classifier	99,8%	PlantVillage 86.198 Images/ 57 Class Controlled Environment	Diverse	RGB	Top View	TensorFlow	Transfer Learning & Fine Tune
				VGG16/LDA		99,0%	Flavia 1.600 Images/ 32 Class Controlled Environment					
				AlexNet/LDA		96,20%	UCI Leaf 443 Images/ 40 Class Real Environment					
				CNN Custom		98,8%	Swedish 1.125 Images/ 15 Class Controlled Environment					
S54	Lin et al. (2019)	2019	3	Fourier Dense Network (FDN)	Multiclass Classifier	92,85%	VOAI UAV 2.293 Images/ 12 Class Real Environment	Diverse	RGB	Aerial View	Caffe	New Architecture
S55	Suryawati et al. (2018)	2018	4,5	VGG	Multiclass Classifier	95,24%	PlantVillage Subset 18.160 Images/ 10 Class Controlled Environment	Tomato	RGB	Top View	TensorFlow	Transfer Learning
S56	Gandhi et al. (2018)	2018	4,5	InceptionV3 MobileNet	Multiclass Classifier	88,6% 92,0%	PlantVillage 54.306 Images/ 38 Class Controlled Environment	Diverse	RGB	Top View	TensorFlow	Data Augmentation
S57	Too et al. (2018)	2018	4,5	VGG InceptionV4	Multiclass Classifier	98,08% 81,33%		Diverse	RGB	Top View	Theano	Fine Tune & Hyperparameters

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Table A.3 (continued)

ID	Refs	Year	IQ	Algorithm	Task of Model	Best Accuracy	Dataset Characteristics	Type of Crops	Type of Images	View Point of Images	Language & Framework	Predominant Approach
S58	de Luna et al. (2018)	2018	4,5	ResNet-50 ResNet-101 ResNet-121 ResNet-152 AlexNet	Multiclass Classifier	99,59% 99,66% 99,59% 99,75% 95,75%	PlantVillage 54.306 Images/ 38 Class Controlled Environment	Tomato	RGB	Top View	MatLab	Layers Custom
S59	Barbedo (2018)	2018	4	GoogLeNet	Multiclass Classifier	87,0%	Customized 4.923 Images/ 4 Class Controlled Environment PDDB Embrapa 1.383 Images/ 56 Class Controlled Environment & Real	Diverse	RGB	Top View	TensorFlow/Keras	Transfer Learning & Data Augmentation
S60	Brahimi et al. (2018)	2018	4	AleNet DenseNet InceptionV3 ResNet SqueezeNet VGG	Multiclass Classifier	99,24% 99,72% 99,76% 99,67% 99,2% 99,49%	PlantVillage 55.038 Images/ 39 Class Controlled Environment	Diverse	RGB	Top View	PyTorch	Transfer Learning & Visualization Maps
S61	Sardogan et al. (2018)	2018	4	Based LeNet + LVQ	Multiclass Classifier	86,0%	PlantVillage Subset 500 Images/ 5 Class Controlled Environment	Tomato	RGB	Top View	TensorFlow	New Architecture
S62	Ferentinos (2018)	2018	4	AlexNet GoogLeNet VGG	Multiclass Classifier	99,44% 97,7% 99,48%	PlantVillage 87.848 Images/ 58 Class Controlled Environment	Diverse	RGB	Top View	TensorFlow	Transfer Learning
S63	Li et al. (2018)	2018	4	CNN Generative Adversario Networks DCGNAS	Binary Classifier Generator/ Discriminator	89,93%	PlantVillage 54.306 Images/ 38 Class Controlled Environment	Diverse	RGB	Top View	TensorFlow	Unsupervised Learning
S64	Ocampo and Dadios (2018)	2018	4	MobileNet	Multiclass Classifier	89,0%	Customized & PDDB 6.970 Images/ 6 Class Controlled Environment & Real	Diverse	RGB	Front View	TensorFlow	Fine Tune & Hyperparameters & Data Augmentation
S65	Cap et al. (2018)	2018	4	Baseado LeNet	Multiclass Classifier	78,0%	Saitama Research Center 960.000 Images/ 2 Class Controlled Environment & Real	Cucumbers	RGB	Front View	TensorFlow	New Architecture
S66	Pardede et al. (2018)	2018	3,5	AutoEnconder CNN	Binary Classifier Generator/ Discriminator	87,01% 80,42	PlantVillage Subset 6.004 Images/ 7 Class Controlled Environment	Potato Corn	RGB	Top View	TensorFlow	Unsupervised Learning
S67	Zhang et al. (2018)	2018	3,5	ResNet (SGD)	Multiclass Classifier	96,51%	PlantVillage Subset 5.550 Images/ 8 Class Controlled Environment	Tomato	RGB	Top View	MatLab	Fine Tune & Hyperparameters & Data Augmentation
S68	Wang et al. (2018)	2018	3,5	Based LeNet	Multiclass Classifier	90,84%	PlantVillage Subset (4.123) Customized (2.430) 6.553 Images/ 8 Class Controlled Environment & Real	Rice Cucumbers	RGB	Top View	TensorFlow	New Architecture
S69	Singh (2018)	2018	3,5	InceptionV3	Multiclass Classifier	92,7% 82,7%	PantVillage 54.306 Images/ 38 Class Controlled Environment Customized 196 Images/ 4 Class Real Environment	Diverse Mango	RGB	Top View	TensorFlow	Transfer Learning

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Table A.3 (continued)

ID	Refs	Year	IQ	Algorithm	Task of Model	Best Accuracy	Dataset Characteristics	Type of Crops	Type of Images	View Point of Images	Language & Framework	Predominant Approach
						92,8%	Customized 811 Images/ 3 Class Real Environment	Peanut				
S70	Wang et al. (2018)	2018	3,5	OR-AC-GAN	Binary Segmentation	98,03%	Customized 60 Images/ 2 Class Controlled Environment	Peppers	Hyperspectral	Top View	TensorFlow	New Architecture
S71	Zhang et al. (2018)	2018	3,5	GoogLeNet	Multiclass Classifier	98,9%	Customized 500 Images/ 9 Class Controlled Environment	Corn	RGB	Front View	Caffe	Layers Custom
S72	Kobayashi et al. (2018)	2018	3,5	InceptionV3	Multiclass Classifier	93,0%	PlantVillage Subset 9.568 Images/ 12 Class Controlled Environment	Apple	RGB	Top View	TensorFlow/Keras	Transfer Learning & Data Augmentation
S73	Barbedo (2018)	2018	3,5	GoogLeNet	Multiclass Classifier	87,0%	PlantVillage Subset & Customized 10.441 Images/ 10 Class Controlled Environment	Corn	RGB	Top View	MatLab	Transfer Learning
S74	Park et al. (2018)	2018	3,5	VGG	Multiclass Classifier	96,0%	& Real Customized 2.639 Images/ 4 Class Controlled Environment	Apple	Hyperspectral	Top View	TensorFlow/Keras	Layers Custom
S75	Fuentes et al. ((2018) 1162.)	2018	3,5	Refinement Filter Bank Based ResNet	Object Detection	96,0%	Customized 8.927 Images/ 10 Class Controlled Environment	Tomato	RGB	Front View	Caffe	New Architecture
S76	Yuan et al. (2018)	2018	3	AlexNet	Multiclass Classifier	95,93%	& Real PlantVillage 54.306 Images/ 38 Class Controlled Environment	Diverse	RGB	Top View	TensorFlow	Transfer Learning
S77	Picon et al. (2018)	2018	3	ResNet	Multiclass Classifier	96,0%	Customized 8.178 Images/ 4 Class Real Environment	Wheat	RGB	Front View	TensorFlow	Layers Custom & Image Segmentation
S78	Tm et al. (2018)	2018	3	LeNet	Multiclass Classifier	94,85%	PlantVillage Subset 18.160 Images/ 10 Class Controlled Environment	Tomato	RGB	Top View	TensorFlow/Keras	Layers Custom
S79	Yang et al. (2018)	2018	3	ResNet (PENLU)	Multiclass Classifier	98,3%	Customized 960 Images/ 3 Class Real Environment	Citrus	RGB	Top View	Caffe	Layers Custom
S80	Jin et al. (2018)	2018	3	R-CNN	Binary Segmentation	95,0%	Customized 10.784 Images Controlled Environment	Corn	Hyperspectral	Top View	Caffe	Image Segmentation
S81	Wallelign et al. (2018)	2018	3	LeNet	Multiclass Classifier	99,32%	PlantVillage Subset 12.673 Images/ 4 Class Controlled Environment	Soybean	RGB	Top View	TensorFlow	Layers Custom
S82	Dang et al. (2018)	2018	3	GoogLeNet	Multiclass Classifier	90,0%	Customized 40 Images 1500 ROI/ 3 Class Controlled Environment	Radish	RGB	Aerial View	PyTorch	Fine Tune & Hyperparameters & Image Segmentation
S83	Ren et al. (2018)	2018	3	VGG	Multiclass Classifier	99,04%	Pest Image DataSet 32.000 Images/ 32 Class Controlled Environment	Diverse	RGB	Front View	TensorFlow/Keras	Layers Custom
S84	Ghosal et al. (2018)	2018	3	CNN xPLNet Based LeNet	Multiclass Classifier	95,04%	Customized 25.000 Images/ 7 Class Controlled Environment	Soybean	RGB	Top View	PyTorch	New Architecture
S85	Khan et al. (2018)	2018	3	VGG		98,6%			Multispectral	Front View	Caffe	

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Table A.3 (continued)

ID	Refs	Year	IQ	Algorithm	Task of Model	Best Accuracy	Dataset Characteristics	Type of Crops	Type of Images	View Point of Images	Language & Framework	Predominant Approach
					Multiclass Classifier		PlantVillage Subset CASC-IFW 6.309 Images/ 6 Class Controlled Environment	Banana & Apple				Transfer Learning & Fine Tune
S86	Zeng et al. (2018)	2018	3	HoResNet Based ResNet	Multiclass Classifier	91,79%	PlantVillage Subset 10.478 Images/ 6 Class Controlled Environment	Tomato	RGB	Top View	TensorFlow	New Architecture
						90,14%	AES-CD9214 9.214 Images/ 6 Class Controlled Environment		Rice			
S87	Alfarisy et al. (2018)	2018	3	CaffeNet	Multiclass Classifier	87,0%	Customized 4.511 Images/ 13 Class Real Environment	Rice	RGB	Front View	Caffe	Data Augmentation
S88	Lim and Estuar (2018)	2018	3	MobileNet	Binary Classifier	65,93%	Customized Fungos 122 Images/ 2 Class Controlled Environment	Banana	Microscopic	Top View	TensorFlow/Keras	Fine Tune & Hyperparameters & Data Augmentation
S89	Ma et al. (2018)	2018	3	AlexNet	Multiclass Classifier	93,4%	Customized 1.184 Images/ 5 Class Controlled Environment	Cucumbers	RGB	Top View	MatLab	Image Segmentation & Data Augmentation
S90	Atole and Park (2018)	2018	3	AlexNet	Multiclass Classifier	91,23%	Customized 600 Images/ 3 Class Controlled Environment	Rice	RGB	Front View	MatLab	Transfer Learning
S91	Rahman et al. (2018)	2018	3	Based VGG16 Stacked Mobile	Multiclass Classifier	95,0%	Customized 1.426 Images/ 9 Class Real Environment	Rice	RGB	Front View	TensorFlow/Keras	New Architecture
S92	Oppenheim et al. (2019)	2018	3	VGG	Multiclass Classifier	96,0%	Customized 493 Images/ 5 Class Controlled Environment	Potato	RGB	Front View	MatLab	Fine Tune & Hyperparameters
S93	Rangarajan et al. (2018)	2018	3	AlexNet VGG16	Multiclass Classifier	97,49% 97,23%	PlantVillage Subset 13.262 Images/ 6 Class Controlled Environment	Tomato	RGB	Top View	MatLab	Transfer Learning & Fine Tune
S94	Lee et al. (2018)	2018	3	VGG & F-RCNN	Object Detection	69,79%	TRES Taiwan Tea Research 1.269 Images/ 3 Class Controlled Environment	Tear Leaf	RGB	Top View	Caffe	Image Segmentation
S95	Li et al. (2018)	2018	3	CaspNet	Binary Classifier	N/A	Customized UAV 500 Images/ 2 Class Controlled Environment	Rice	RGB	Aerial View	MatLab	Image Segmentation
S96	Wang and Zhang (2018)	2018	3	FCN AlexNet	Multiclass Segmentation	96,0%	Customized 750 Images/ 6 Class Controlled Environment	Corn	RGB	Top View	MatLab	Fine Tune & Image Segmentation
S97	Kerkech et al. (2018)	2018	3	LeNet-5	Multiclass Classifier	95,8%	Customized UAV 70.560 Images/ 4 Class Controlled Environment	Grape	RGB	Aerial View	MatLab	Image Segmentation
S98	Fujita et al. (2018)	2018	3	VGG	Multiclass Classifier	93,6%	Customized 9.000 Images/ 9 Class Controlled Environment	Cucumbers	RGB	Front View	Caffe	Transfer Learning
S99	Durmus et al. (2017)	2017	4,5	AlexNet SqueezeNet	Multiclass Classifier	95,65% 94,3%	PlantVillage 54.306 Images/ 38 Class Controlled Environment	Tomato	RGB	Top View	Caffe	Transfer Learning
S100	Lu et al. (2017)	2017	4	Based LeNet	Multiclass Classifier	95,48%		Rice	RGB	Front View	PyTorch	New Architecture

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Table A.3 (continued)

ID	Refs	Year	IQ	Algorithm	Task of Model	Best Accuracy	Dataset Characteristics	Type of Crops	Type of Images	View Point of Images	Language & Framework	Predominant Approach
S101	Wang et al. (2017)	2017	4	VGG	Multiclass Classifier	90,4%	Customized 500 Images/ 10 Class Controlled Environment PlantVillage Subset 2.086 Images/ 4 Class Controlled Environment	Apple	RGB	Top View	Theano	Fine Tune & Hyperparameters
S102	Fuentes et al. (2017)	2017	4	R-FCN & ResNet	Object Detection	85,98%	Customized 5.000 Images/ 9 Class Controlled Environment	Tomato	RGB	Front View	TensorFlow/Keras	Data Augmentation
S103	DeChant et al. (2017)	2017	4	Based AlexNet	Binary Classifier	97,8%	Customized 1.796 Images/ 2 Class Controlled Environment	Corn	RGB	Front View	Theano	Image Segmentation
S104	Park et al. (2017)	2017	3,5	LeNet	Multiclass Classifier	89,7%	Customized 1500 Images/ 4 Class Controlled Environment	Strawberry	RGB	Front View	TensorFlow	Layers Custom
S105	Atabay (2017)	2017	3,5	ResNet	Multiclass Classifier	97,57%	PlantVillage Subset 19.742 Images/ 10 Class Controlled Environment	Tomato	RGB	Top View	Caffe	Layers Custom
S106	Ramcharan et al. (2017)	2017	3,5	Inception V3	Multiclass Classifier	98,0%	Original Cassava (2.756) Leaf Cassava (15.000) Controlled Environment	Cassava	RGB	Front View	TensorFlow	Transfer Learning
S107	Peng and Cai (2017)	2017	3,5	FCN	Binary Segmentation	87.5%	& Real Customized 546 Images/ 2 Class Controlled Environment	Apple	RGB	Top View	Caffe	Image Segmentation
S108	Yamamoto et al. (2017)	2017	3	AlexNet	Multiclass Classifier	78%	PlantVillage Subset 18.149 Images/ 9 Class Controlled Environment	Tomato	RGB	Top View	MatLab	Fine Tune & Hyperparameters
S109	Šulc and Matas (2017)	2017	3	Inception-ResNet V2	Multiclass Classifier	98,8%	Middle European Wood 9.745 Images/ 153 Class Controlled Environment	Diverse	RGB	Top View	PyTorch	Fine Tune & Layers Custom
						97,1%	Austrian Federal Forest (AFF) 134 Images/ 5 Class Controlled Environment					
						99,4%	Flavia leaf Dataset 1907 Images/ 32 Class Controlled Environment					
						99,2%	Foliage Leaf Dataset 6.000 Images/ 60 Class Controlled Environment					
						99,7%	Swedish Leaf Dataset 1.125 Images/ 15 Class Controlled Environment					
						81,2%	Leafsnap Dataset 23.147 Images/ 185 Class Controlled Environment					
S110	Cortes (2017)	2017	3	LeNet & GANs		78%		Diverse	RGB	Top View	TensorFlow	Layers Custom

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Table A.3 (continued)

ID	Refs	Year	IQ	Algorithm	Task of Model	Best Accuracy	Dataset Characteristics	Type of Crops	Type of Images	View Point of Images	Language & Framework	Predominant Approach
S111	Sun et al. (2017 (2017))	2017	3	LeNet	Multiclass Classifier	89,6%	PlantVillage Segmentado 86.147 Images/ 57 Class Controlled Environment	Flowers	RGB	Top View	TensorFlow/Keras	Layers Custom
S112	Ha et al. (2017)	2017	3	VGG	Multiclass Classifier	97,4%	BjfuGloxinia 321 Images/ 3 Class Controlled Environment	Radish	RGB	Aerial View	Caffe	Layers Custom
S113	Liu et al. (2017)	2017	3	Based AlexNet	Multiclass Classifier	97,62%	Customized 1.734 Images/ 4 Class Real Environment	Apple	RGB	Top View	Caffe	New Architecture
S114	Amara et al. (2017)	2017	3	Based LeNet	Multiclass Classifier	92,88%	Customized 13.689 Images/ 4 Class Controlled Environment	Banana	RGB	Top View	MatLab	New Architecture
S115	Sladojevic et al. (2016 (2016))	2016	4,5	CaffeNet	Multiclass Classifier	96,3%	PlantVillage Subset 3.700 Images/ 3 Class Controlled Environment	Diverse	RGB	Front View	Caffe	Fine Tune & Hyperparameters
S116	Tan et al. (2016)	2016	4	Based LeNet	Multiclass Classifier	95,5%	Customized 400 Images Controlled Environment	Apple	RGB	Top View	MatLab	New Architecture
S117	Mohanty et al. (2016)	2016	3,5	AlexNet GoogLeNet	Multiclass Classifier	99,27% 97,24%	PlantVillage 54.306 Images/ 38 Class Controlled Environment	Diverse	RGB	Front View	Caffe	Transfer Learning
S118	Nachtigall et al. (2016)	2016	3	AlexNet	Multiclass Classifier	97,3%	TreeApple Embrapa 1.450 Images/ 3 Class Controlled Environment	Apple	RGB	Top View	TensorFlow/Keras	Fine Tune & Hyperparameters
S119	Fujita et al. (2016)	2016	3	VGG	Multiclass Classifier	82,9%	Saitama Research Center 7,320 Images/ 4 Class Controlled Environment	Cucumbers	RGB	Top View	Caffe	Layers Custom & Data Augmentation
S120	Atabay (2016)	2016	3	AlexNet Based AlexNet	Multiclass Classifier	75,05% 97,24%	Customized 7.520 Images/ 4 Class Controlled Environment & Real Flavia 1.907 Images/ 32 Class Controlled Environment	Diverse	RGB	Top View	MatLab	New Architecture
S121	Kawasaki et al. (2015)	2015	4,5	CaffeNet	Multiclass Classifier	99,11% 94,9%	Swedish 75 Images/ 15 Class Controlled Environment Customized 800 Images/ 3 Class Controlled Environment	Cucumbers	RGB	Top View	Caffe	Data Augmentation

Appendix B. Summarization

Table B.4.

Table B.4

Diseases and their Pathogens grouped by Dataset and Studies

ID Study	DataSet Name	Type Crop	Disease Name	Pathogen
S3, S4, S9, S11, S13, S14, S18, S21, S23, S26, S27, S28, S29, S35, S41, S48, S51, S53, S55, S56, S57, S60, S61, S62, S63, S65, S66, S67, S68, S69, S72, S73, S76, S78, S81, S85, S86, S93, S99, S101, S105, S108, S110, S114, S117	PlantVillage	Apple	Apple scab Black rot Cedar apple rust Leaf Black Rot	Fungi Fungi Fungi Fungi
		Cherry	Powdery mildew	Fungi
		Corn	Cercospora leaf spot Gray leaf spot Common rust	Fungi Fungi Fungi
		Grape	Northen leaf blight Black rot Esca (Black measles)	Fungi Fungi Fungi
		Orange	Bacterial leaf spot Isariopsis leaf spot Haunglongbing (Citrus greening)	Bacterial Fungi Bacterial
		Peach	Bacterial spot	Bacterial
		Bell Pepper	Bacterial spot	Bacterial
		Potato	Early blight Late blight	Fungi Fungi
		Squash	Powdery mildew	Fungi
		Strawberry	Leaf scorch Bacterial spot	Fungi Bacterial
		Tomato	Early blight Late blight Leaf mold Septoria leaf spot Spider mites Two spotted Target spot	Fungi Fungi Fungi Fungi Plague Fungi
			Tomato mosaic virus Tomato yellow leaf curl virus	Virus Virus
		Mango	Anthracnose	Fungi
		Banana	Banana Sigatoka Banana Cordana musae	Fungi Fungi
			Deightoniella Leaf Spot	Fungi
			Banana Diamond Leaf Spot	Fungi
S1	Custom	Cucumbers	Downy mildew Anthracnose Gray mold Angular leaf spot Black spot Powdery mildew	Fungi Fungi Fungi Bacterial Fungi Fungi
S2	Custom Spike	Wheat	Nutritional Deficiency	Abiotic
S3	Custom	Tomato	Bacterial spot Early blight Late blight Leaf mold Spider mites Two spotted Target spot Tomato yellow leaf curl virus	Bacterial Fungi Fungi Fungi Plague Fungi Virus
S5, S19, S42	Crop Disease AI Challenger 2018	Apple	Apple scab Apple Frogeye Spot Cedar Apple Rust	Fungi Fungi Fungi
		Cherry	Powdery mildew	Fungi
		Corn	Cercospora zeaemaydis	Fungi

Table B.4 (continued)

ID Study	DataSet Name	Type Crop	Disease Name	Pathogen	
S6	Custom	Grape	Puccinia polysora	Fungi	
			Corn Curvularia leaf spot	Fungi	
			Dwarf mosaic virus	Virus	
			Black Rot	Fungi	
			Black Measles	Fungi	
			Leaf Blight	Fungi	
			Greening	Bacterial	
			Peach	Bacterial spot	Bacterial
			Pepper	Pepper Scab	Bacterial
			Potato	Early blight	Fungi
		Strawberry	Late blight	Fungi	
			Strawberry Scorch	Fungi	
			Powdery mildew	Fungi	
			Bacterial spot	Bacterial	
			Early blight	Fungi	
			Late blight	Fungi	
			Leaf mold	Fungi	
			Target spot	Fungi	
			Septoria leaf spot	Fungi	
			Spider mites Two spotted	Plague	
		Winter Wheat	YLCV Virus	Virus	
			Tomato mosaic virus	Virus	
			Septoria Tritic	Fungi	
			Puccinia striiformis	Fungi	
			Puccinia recondita	Fungi	
			Septoria nodorum	Fungi	
			Drechslera tritici-repentis	Fungi	
			Oculimacula	Fungi	
			Yallundae	Fungi	
			Gibberella zeae	Fungi	
Corn	Blumeria graminis	Fungi			
	Helminthosporium turcicum	Fungi			
	Rape Seed	Phoma lingam	Fungi		
	Winter Barley	Pyrenophora teres	Fungi		
		Ramularia collo-cygni	Fungi		
		Rhynchosporium secalis	Fungi		
	Commom Rice	Puccinia hordei	Fungi		
		Thanatephorus cucumeris	Fungi		
	S7	Custom	Soybean	Pyricularia oryzae	Fungi
	S8	Custom	Corn	Charcoal rot	Fungi
S9	Custom	Cucumbers	Northen leaf blight	Fungi	
			Scab Angular	Fungi	
			Powdery mildew	Fungi	
			Downy mildew	Fungi	
			Anthracnose and Scab	Fungi	
			Scab	Fungi	
			Magnaporthe grisea	Fungi	
			S10	Custom	Commom Rice
S12	Custom	Winter Wheat	Powdery mildew	Fungi	
			Cochliobolus heterostrophus	Fungi	
			Bacterial Leaf Streak	Bacterial	
			Leaf Blight	Fungi	
S15 S16,S59	Custom PDDB Embrapa	Apple	Stripe Rust	Fungi	
			Leaf Rust	Fungi	
			Anthracnose	Fungi	
			Pineapple Fruit Borer	Plague	
		Cotton	Fusarirose	Fungi	
			Black Rot	Fungi	
			Fusarium Wilt	Fungi	
			Myrothecium Leaf Spot	Fungi	
			Soreshin	Fungi	
			Areolate Mildew	Fungi	
Rice	Rice Blast	Fungi			
	Leaf Scald	Fungi			
Coffe		Leaf Miner	Plague		

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Table B.4 (continued)

ID Study	DataSet Name	Type Crop	Disease Name	Pathogen
			Cercospora leaf spot	Fungi
			Rust	Fungi
			Bacterial Blight	Bacterial
			Blister Spot	Fungi
			Brown Leaf Spot	Fungi
			Pellicularia	Fungi
			Koleroga	
			Damping Off	Fungi
		Cashew Tree	Algae	Algae
			Anthrachnose	Fungi
			Worm	Abiotic
			Angular leaf spot	Bacterial
			Black Mould	Fungi
			Powdery mildew	Fungi
			Gummosis	Bacterial
			Seedling Blight	Fungi
			Phoma costaricensis	Fungi
		Sugarcane	Rust	Fungi
			Ring Spot	Fungi
			White Fly	Plague
			Red Stripe	Bacterial
		Citrus	Algae	Algae
			Alternaria Brown Spot	Fungi
			Citrus Canker	Bacterial
			Citrus Variedgated	Bacterial
			Chlorosis	
			Nutritional Deficiency	Abiotic
			Felt	Fungi
			Sooty Mold	Fungi
			Leprosis	Virus
			Halo Blight	Bacterial
			Citrus Greasy Spot	Fungi
			Black Spot	Fungi
			Mosaic	Virus
			Penicillium	Fungi
			Postbloom Fruit	Fungi
			Drop of Citrus	
			White-Thread Blight	Fungi
			Scab	Fungi
		Coconut Tree	Steneotarsonemus furcatus	Plague
			Aceria guerreronis	Plague
			Caterpillar damage	Plague
			Coconut Scale	Plague
			Phytotoxicity	Abiotic
			Sooty Mold	Fungi
			Dreschlera incurvata	Fungi
			Sphaerodothis acrocomiae	Fungi
			Phyllachora torrendiella	Fungi
			Cylindrocladium leaf spot	Fungi
			White Fly	Plague
			Coconut Rot	Fungi
			Premature Fruit	Abiotic
			Drop	
			Coconut Leaf Blight	Fungi
			Stem Bleeding	Fungi
		Cabbage	Alternaria Leaf Spot	Fungi
			Powdery mildew	Fungi
		Cupuaçu	Lasioidiplodia Rot	Fungi
			Witches Broom	Fungi
		Dry Bean	Anthrachnose	Fungi
			Cercospora leaf spot	Fungi
			Cowpea mild mottle virus	Virus
			Common Bacterial Blight	Bacterial
			Rust	Fungi
			Phytotoxicity	Abiotic
			Hedylepta Indicata	Plague

Table B.4 (continued)

ID Study	DataSet Name	Type Crop	Disease Name	Pathogen
			Target Leaf Spot	Fungi
			Bacterial Spot	Bacterial
			Web Blight	Fungi
			Powdery mildew	Fungi
			Charcoal Rot	Fungi
			Bean golden mosaic	Virus
		Papaya	Anthrachnose	Fungi
			Mosaic Virus	Virus
			Smallpox	Virus
		Cassava	Green Mite	Plague
			Anthrachnose	Fungi
			Bacterial Blight	Bacterial
			White Leaf Spot	Fungi
			Brown Leaf Spot	Fungi
			Cassava Common Mosaic	Virus
			Leaf Galls	Plague
			Powdery mildew	Fungi
			Root Rot	Fungi
			Blight Leaf Spot	Fungi
		Passion Fruit	Cercospora leaf spot	Fungi
			Scab	Fungi
			Fusariosis	Fungi
			Dione junio junio	Plague
			Bacterial Spot	Bacterial
			Mosaic Virus	Virus
			Base Rot	Fungi
			Senescence	Abiotic
			Septoria leaf spot	Fungi
			Woodiness virus	Virus
		Melon	Bacterial Fruit Blotch	Bacterial
			Mosaic	Virus
			Meloidogyne sp.	Nematode
			Powdery mildew	Fungi
			Anthrachnose	Fungi
			Bushy Stunt	Bacterial
			Tropical Rust	Fungi
			Southern Corn Rust	Fungi
			Scab	Fungi
			Southern Corn Leaf Blight	Fungi
			Phaeosphaeria Leaf Spot	Fungi
			Diplodia Leaf Streak	Fungi
			Physoderma Brown Spot	Fungi
			Northern leaf blight	Fungi
			Red Ring - Bursaphelenchus cocophilus	Nematode
			Anthrachnose	Fungi
			Fusarium Wilt	Fungi
			Necrotic Spotting	Fungi
			Curvularia Leaf Spot	Fungi
			Black Sooty Mold	Fungi
			Thielaviopsis rot	Fungi
			Anthrachnose	Fungi
			Yellow Wilt	Fungi
			Phytophthora Foot Rot	Fungi
			White-Thread Blight	Fungi
			Bacterial Blight	Bacterial
		Soybean	Cercospora Blight	Fungi
			Charcoal Rot	Fungi
			Rust	Fungi
			Rust and Target Spot	Fungi
			Copper	Abiotic
			Phytotoxicity	
			Soybean mosaic virus	Virus
			Southern Blight	Fungi
			Myrothecium Leaf Spot	Fungi
				Fungi

(continued on next page)

Table B.4 (continued)

ID Study	DataSet Name	Type Crop	Disease Name	Pathogen
S18	Custom	Wheat	Rhizoctonia aerial blight	
			Downy mildew	Fungi
			Southern Blight	Fungi
			Powdery mildew	Fungi
			Powdery mildew and Rust	Fungi
			Phytophthora Rot	Fungi
			Brown Leaf Spot	Fungi
			Wheat Blast	Fungi
			Rust	Fungi
			Tan Spot	Fungi
			Powdery mildew	Fungi
			Anthraxnose	Fungi
			Bacterial Canker	Bacterial
			Rust	Fungi
			Isariopsis Leaf Spot	Fungi
			Downy mildew	Fungi
			Powdery mildew	Fungi
			Grapevine fanleaf virus	Virus
S20	Custom	Banana	Anthraxnose	Fungi
S22	Custom	Wheat	Xanthomonas wilt	Bacterial
			Bunchy top disease	Virus
			Black sigatoka	Fungi
			Yellow sigatoka	Fungi
			Fusarium wilt	Fungi
			Corm weevil	Plague
			Dried/old leaves	Abiotic
			Yellow Rust	Fungi
			Esca (Black measles)	Fungi
			Apple Scab	Fungi
S24	Custom	Grape	Black Rot	Fungi
S25	Custom	Apple	Chlorosis	Abiotic
S28	Custom	Grape	Esca (Black measles)	Fungi
			Coryneum	Fungi
			beijerinckii	
			Peach	Plague
			sphaerolectanium prunastri	
			Xanthomonas arboricola	Bacterial
			Erwinia amylovora	Bacterial
			Monilinia laxa	Fungi
			Apricot monilia laxa	Fungi
			Walnut leaf Gall	Plague
S31	BU-Guava-Leaf	Cherry	Mite	
			Cherry myzus cerasi	Plague
			Algal leaf spot	Alga
			Rust	Fungi
			White Fly	Plague
			Alternaria Leaf Spot	Fungi
			Brown Leaf Spot	Fungi
			Mosaic	Virus
			Grey Spot	Fungi
			Rust	Fungi
S33	Custom	Cucumbers	Powdery mildew	Fungi
S34	Custom	Citrus	Pezothrips	Plague
			Parlatoria pergandii	Plague
			Aonidiella aurantii	Plague
			Scirtothrips	Plague
			Venturia inaequalis	Fungi
			Gymnosporangium juniperi-virginianae	Fungi
			Podosphaera leucotricha	Fungi
			Botryosphaeria obtuse	Fungi
			Alternaria pomi	Fungi
			Xanthomonas euvesicatoria	Bacterial
S35	PlantDisease	Apple	Xanthomonas campestris	Bacterial
			Blumeriella jaapii	Fungi
			Podosphaera spp.	Fungi

Table B.4 (continued)

ID Study	DataSet Name	Type Crop	Disease Name	Pathogen
S36	Custom	Grape	Uncinula necator	Fungi
			Plasmopara viticola	Fungi
			Botrytis cinerea	Fungi
			Botryosphaeria obtuse	Fungi
			Pseudocercospora vitis	Fungi
			Guignardia bidwellii	Fungi
			Phaeomoniella spp.	Fungi
			Peronospora destructor	Fungi
S37	Custom	Onion	Clasterosporium carpophilum	Fungi
			Alternaria solani	Fungi
			Polystigma rubrum	Fungi
			Plum Plox	Virus
			Tranzschelia pruni-spinosae	Fungi
			Stigmina carpofilia	Fungi
			Mycosphaerella fragariae	Fungi
S38	Custom	Peach	Cercospora beticola	Fungi
			Phytophthora infestans	Fungi
			Septoria lycopersici	Fungi
			Erysiphe graminis	Fungi
			Puccinia spp.	Fungi
			Septoria spp.	Fungi
			Mildew	Fungi
			Potato PVY	Virus
			Tea bud blight	Fungi
			Tea leaf blight	Fungi
S39	Custom	Tea Leaf	Tea red Scab	Fungi
			Marssonina Blotch	Fungi
			Alternaria Leaf Spot	Fungi
			Brown streak	Virus
			Mosaic	Virus
			Green Mite	Plague
			Red Mite	Plague
			Brown Leaf Spot	Fungi
			Nutritional Deficiency	Abiotic
S40	IITA Cassava	Cassava	Northern leaf blight	Fungi
			Puccinia Sorghi	Fungi
			Gray Leaf Spot	Fungi
			Malformed fruit	Abiotic
			Blotchy ripening	Abiotic
			Puffy fruit	Abiotic
			Dehiscent fruit	Abiotic
			Blossom-end rot	Abiotic
			Sunscald	Abiotic
			Tomato Virus	Virus
S43	Custom	Corn	Gray mold	Fungi
			Tomato ulcer	Fungi
			Anthraxnose	Fungi
			Cercospora Leaf Spot	Fungi
S44	Custom	Tomato	White Leaf Spot	Fungi
			Birds eye spot	Fungi
			Red leaf spot	Alga
			Gray blight	Fungi
			Anthraxnose	Fungi
			Brown Blight	Bacterial
			Algal leaf spot	Alga
			Northern leaf blight	Fungi
			Citrus Canker	Bacterial
			Ceratitidis capitata	Plague
S45	Custom	Sugar beets	Diaphorina citri	Plague
			Kuwayama	
			Anoplophora chinensis	Plague
			Halyomorpha halys	Plague
			Nezara viridula	Plague
			Othreis fullonica	Plague
			Papilio demodocus	Plague
S46	Custom	Tea Leaf		
S47	Custom	Citrus		
S49	Custom	Citrus		
S50	Custom	Citrus		

(continued on next page)

Table B.4 (continued)

ID Study	DataSet Name	Type Crop	Disease Name	Pathogen
S52	Custom	Tea Leaf	Metcalfa pruinosa (Say)	Plague
			Planococcus citri	Plague
			Toxoptera citricida	Plague
			Hemiptera: Coccidae	Plague
			Thaumatotibia leucotreta	Plague
			Root Weevil	Plague
			Scudderella furcata	Plague
			Cicadoidea	Plague
			Cornu aspersum	Plague
			Homalodisca vitripennis	Plague
			Colletotrichum gloeosporioides	Fungi
			Xanthomonas axonopodis	Bacterial
			Diaporthe citri	Fungi
			Elsinoë fawcettii	Fungi
			Liriomyza brassicae	Plague
			Capnodium spp	Fungi
			Locusta migratoria	Plague
			Parasa lepida	Plague
			Gypsy moth larva	Plague
			Empoasca flavescens	Plague
			Spodoptera exigua	Plague
			Chrysoschus chinensis	Plague
			Laspeyresia pomonella larva	Plague
			Spodoptera exigua larva	Plague
			Atractomorpha sinensis	Plague
			Laspeyresia pomonella	Plague
S53, S109, S120	Flavia Dataset	Phyllostachys edulis (Carr.) Houz.	N/A	N/A
			Aesculus chinensis	N/A
			Berberis anhwensis	N/A
			Ahrendt	N/A
			Cercis chinensis	N/A
			Indigofera tinctoria L.	N/A
			Acer Palmatum	N/A
			Phoebe nanmu	N/A
			Kalopanax septemlobus	N/A
			Cinnamomum japonicum Sieb.	N/A
			Koelreuteria paniculata Laxm.	N/A
			Ilex macrocarpa Oliv.	N/A
			Pittosporum tobira (Thunb.) Ait. f.	N/A
			Chimonanthus praecox L.	N/A
			Cinnamomum camphora	N/A
			Viburnum awabuki K. Koch	N/A
			Osmanthus fragrans Lour.	N/A
			Cedrus deodara (Roxb.) G. Don	N/A

Table B.4 (continued)

ID Study	DataSet Name	Type Crop	Disease Name	Pathogen
	UCI Leaf Dataset	Ginkgo biloba L.	N/A	N/A
		Lagerstroemia indica (L.) Pers.	N/A	N/A
		Nerium oleander L.	N/A	N/A
		Podocarpus macrophyllus	N/A	N/A
		Prunus serrulata Lindl.	N/A	N/A
		Ligustrum lucidum Ait. f.	N/A	N/A
		Tonna sinensis M. Roem.	N/A	N/A
		Prunus persica (L.) Batsch	N/A	N/A
		Manglietia fordiana Oliv.	N/A	N/A
		Acer buergerianum Miq.	N/A	N/A
		Mahonia bealei (Fortune) Carr.	N/A	N/A
		Magnolia grandiflora L.	N/A	N/A
		Populus canadensis	N/A	N/A
		Moench Liriodendron chinense	N/A	N/A
		Citrus reticulata Blanco	N/A	N/A
		Quercus suber	N/A	N/A
		Salix atrocinerea	N/A	N/A
		Populus nigra	N/A	N/A
		Alnus sp	N/A	N/A
		Quercus robur	N/A	N/A
		Crataegus monogyna	N/A	N/A
		Ilex aquifolium	N/A	N/A
		Nerium oleander L.	N/A	N/A
		Betula pubescens	N/A	N/A
		Tilia tomentosa	N/A	N/A
		Acer palmaturu	N/A	N/A
		Celtis sp	N/A	N/A
		Corylus avellana	N/A	N/A
		Castanea sativa	N/A	N/A
		Populus alba	N/A	N/A
		Acer negundo	N/A	N/A
		Taxus bacatta	N/A	N/A
		Polypodium vulgare	N/A	N/A
		Pinus sp	N/A	N/A
		Fraxinus sp	N/A	N/A
		Primula vulgaris	N/A	N/A
		Erodium sp	N/A	N/A
		Bougainvillea sp	N/A	N/A
		Arisarum vulgare	N/A	N/A
		Euonymus japonicus	N/A	N/A
		Ilex perado ssp azorica	N/A	N/A
		Magnolia soulangeana	N/A	N/A
		Buxus sempervirens	N/A	N/A
		Urtica dioica	N/A	N/A

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Table B.4 (continued)

ID Study	DataSet Name	Type Crop	Disease Name	Pathogen	
S54	Swedish Leaf Dataset	Podocarpus sp	N/A	N/A	
		Acca sellowiana	N/A	N/A	
		Hydrangea sp	N/A	N/A	
		Pseudosasa japonica	N/A	N/A	
		Magnolia grandiflora L.	N/A	N/A	
		Geranium sp	N/A	N/A	
		Aesculus californica	N/A	N/A	
		Chelidonium majus	N/A	N/A	
		Ulmus carpinifolia	N/A	N/A	
		Acer	N/A	N/A	
		Salix aurita	N/A	N/A	
		Quercus	N/A	N/A	
		Alnus incana	N/A	N/A	
		Betula pubescens	N/A	N/A	
		Salix alba ‘Sericea’	N/A	N/A	
		Populus tremula	N/A	N/A	
		Ulmus glabra	N/A	N/A	
		Sorbus aucuparia	N/A	N/A	
		Salix sinerea	N/A	N/A	
		Populus	N/A	N/A	
	Tilia	N/A	N/A		
	Sorbus intermedia	N/A	N/A		
	VOAI UAV Dataset	Fagus silvatica	N/A	N/A	
	Adenanthera pavonina Linn	N/A	N/A		
	Blechnum orientale Linn	N/A	N/A		
	Psychotria serpens Linn	N/A	N/A		
	Carmona microphylla (Lam)	N/A	N/A		
	Dicranopteris dichotoma (Thunb)	N/A	N/A		
	Eucalyptus citriodora Hook	N/A	N/A		
	Photinia serrulata Lindl	N/A	N/A		
	Pinus massoniana Lamb	N/A	N/A		
	Withered dicranopteris dichotoma	N/A	N/A		
	Withered Mosla chinensis Maxim	N/A	N/A		
	Miscanthus sinensis Anderss	N/A	N/A		
	Acacia confusa Merr	N/A	N/A		
	S58	Custom	Tomato	Tomato Leaf Miner	Plague
	S64	Custom PDDS	Apple	Phoma Rot	Fungi
				Target Leaf Spot	Fungi
				Scab	Fungi
				Black Rot	Fungi
Corn			Ceder Rust	Fungi	
			Cercospora leaf spot	Fungi	
Tomato			Common rust	Fungi	
			Northen leaf blight	Fungi	
	Early Blight	Fungi			
	Leaf Mold	Fungi			
		Septoria leaf spot	Fungi		

Table B.4 (continued)

ID Study	DataSet Name	Type Crop	Disease Name	Pathogen
S65	Saama Research Center	Cucumbers	Target Leaf Spot N/A	Fungi N/A
S69	Custom	Mango	Bacterial Canker	Bacterial
			Mildew Mango	Fungi
			Phoma Blight	Fungi
			Red Rust	Alga
		Groundnut	Leaf Spot	Fungi
			Bud Necrosis	Virus
S70	Custom	Pepper	Tomato spotted wilt virus	Virus
S71	Custom	Corn	Curvularia Leaf Spot	Fungi
			Dwarf Mosaic	Virus
			Gray Leaf Spot	Fungi
			Northern leaf blight	Fungi
			Brown Spot	Fungi
			Round Spot	Bacterial
			Rust	Fungi
			Southern Blight	Fungi
S74	Custom	Apple	Apple Marssonina blotch (AMB)	Fungi
			Young Leaf	Fungi
			Nutritional Deficiency	Abiotic
S75	Custom	Tomato	Leaf Mold	Fungi
			Gray Mold	Fungi
			Canker	Bacterial
			Plague	N/A
			Miner	Plague
			Powdery mildew	Fungi
			White Fly	Plague
			Yellow leaf curl	Virus
			Nutritional excess	Abiotic
S77	Custom	Wheat	Rust	Fungi
			Septoria leaf spot	Fungi
			Tan Spot	Fungi
S79	Custom	Citrus	Yellowing	N/A
S80	Custom	Corn	N/A	N/A
S82	Custom	Radish	Fusarium wilt	Fungi
S83	Pest Image Dataset	N/A	Anomala corpulenta	Plague
			Motsehlusi	Plague
			Papilio xuthus	Plague
			Linnaeus	
			Gryllotalpa	Plague
			Erthesina fullo	Plague
			Apriona germari	Plague
			Lebeda nobilis	Plague
			Walker	
			Papilio bianor	Plague
			Ctenuchidae	Plague
			Gongylopus adyposus	Plague
			Platyomia piel Kato	Plague
			Stichophthalma louisiana	Plague
			Anoplophora chinensis	Plague
			Eurydema dominulus	Plague
			Pieris rapae crucivora	Plague
			Nezara viridula	Plague
			Linnaeus	Plague
			Parnara guttata	Plague
			Bremer et Grey	Plague
			Dichocrocis punctiferalis	Plague
			Guenée	Plague
			Ostrinia nubilalis	Plague
			Gryllulus	Plague
			Grasshopper	Plague
			Pergesa elpenorlewisi	Plague
			Henosepilachna	Plague

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Table B.4 (continued)

ID Study	DataSet Name	Type Crop	Disease Name	Pathogen
S97 S98	Custom	Grape Cucumbers	Leaf Spot	Fungi
			Brown Spot	Fungi
			Streak and round spot	Virus
			Esca (Black measles)	Fungi
			Melon yellow spot virus	Virus
			Zucchini yellow mosaic virus	Virus
			Cucurbit chlorotic yellows virus	Virus
			Cucumber mosaic virus	Virus
			Papaya ring spot virus	Virus
			Watermelon mosaic virus	Virus
S100	Custom	Rice	Green mottle mosaic Virus	Virus
			Downy mildew	Fungi
			bacterial wilt	Bacterial
			seeding blight	Fungi
			bacterial sheath rot	Bacterial
			Bacterial Leaf Blight	Bacterial
			Sheath Rot	Fungi
			Sheath Blight	Fungi
			bakanae disease	Fungi
			Brown Spot	Fungi
S102	Custom	Tomato	False smut	Fungi
			Rice Blast	Fungi
			Nutritional excess	Abiotic
			Whitefly	Plague
			Powdery mildew	Fungi
			Leaf Miner	Plague
			Canker	Bacterial
			Gray mold	Fungi
			Leaf mold	Fungi
			Northern leaf blight (NLB)	Fungi
S103	Custom	Maize		
S104	Custom	Strawberry	Gray mold rot	Fungi
			Powdery mildew	Fungi
S106	Custom	Cassava	Fusarium wilt	Fungi
			Anthracnos	Fungi
			Cassava mosaic disease	Virus
			Cassava brown streak disease	Virus
S107 S109	Custom Middle European Wood Austrian Federal Forest (AFF) Foliage Leaf Dataset Leafsnap Dataset	Apple Diverse	Green mite damage	Plague
			Red mite damage	Plague
			Brown leaf spot	Fungi
			Abiotic disturb	Abiotic
S111 S112	BjfuGloxinia Custom	Flowers Radish	N/A	N/A
S113	Custom	Apple	N/A	N/A
S115	Custom	Pear	Fusarium wilt of radish	Fungi
			Mosaic	Virus
			Brown spot	Fungi
			Rust	Fungi
		Alternaria leaf spot	Fungi	
		Porosity	NA	
		Erwinia amylovora	Bacterial	
		Venturia	Fungi	
Cherry Peach	Porosity	NA		
	Porosity	NA		
	Powdery mildew	Fungi		
	Taphrina deformans	Fungi		

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Table B.4 (continued)

ID Study	DataSet Name	Type Crop	Disease Name	Pathogen
S116	Custom	Apple	Erwinia amylovora	Bacterial
			Venturia	Fungi
			Powdery mildew	Fungi
		Pair	Rust	Fungi
			Gymnosporangium sabinae	Fungi
			Gray leaf spot	Fungi
		Grapevine	Wilt	Fungi
			Mites	Plague
			Powdery mildew	Fungi
		Apple	Downy mildew	Fungi
			Scab skin	Fungi
			Black rot	Fungi
		Apple	Scar skin	Virus
			Ring spot	Fungi
			Potassium deficiency	Abiotic
S118	Custom	Apple	Magnesium deficiency	NA
			Scab damage	Fungi
			Glomerella stain	Fungi
			Herbicide damage	NA
			Melon yellow spot virus: MYSV	Virus
			Zucchini yellow mosaic virus: ZYMV	Virus
			Cucurbit clorotic yellows virus: CCYV	Virus
			Cucumber mosaic virus: CMV	Virus
			Papaya ring spot virus: PRSV	Virus
			Watermelon mosaic virus: WMV	Virus
			Green mottle mosaic virus	Virus
			Melon yellow spot virus: MYSV	Virus
			Zucchini yellow mosaic virus: ZYMV	Virus
S121	Custom	Cucumber		

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